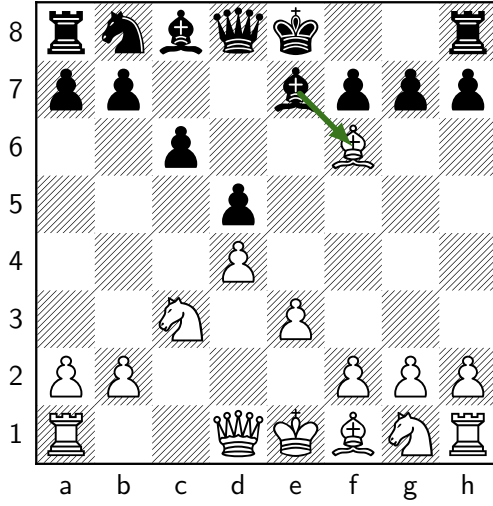


# Melanie-ANALOGY — QGD, Exchange

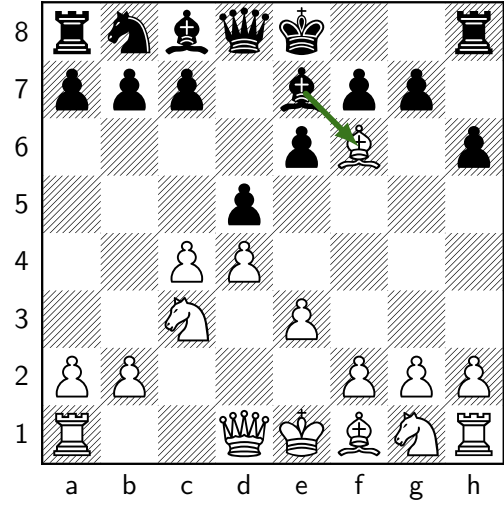
Melanie (Deep\_Vision)

## Abstract

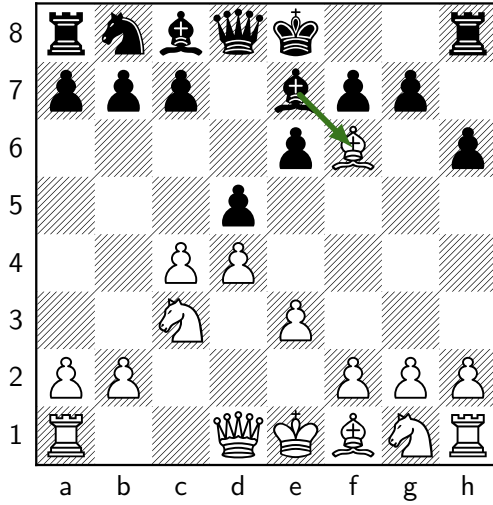
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under  $\mu$  – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



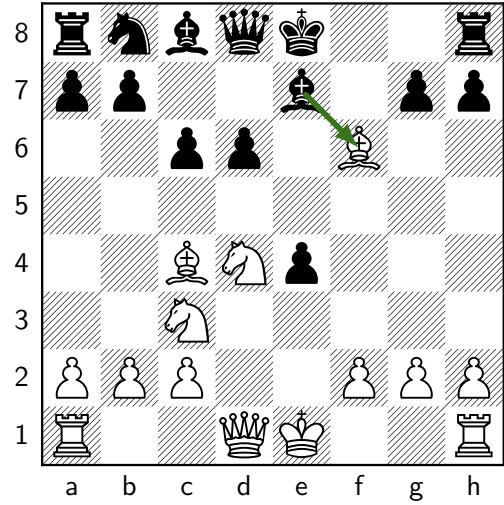
(a)



(b)

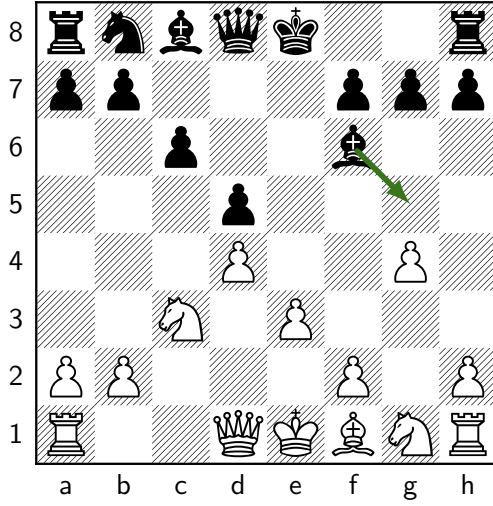


(c)

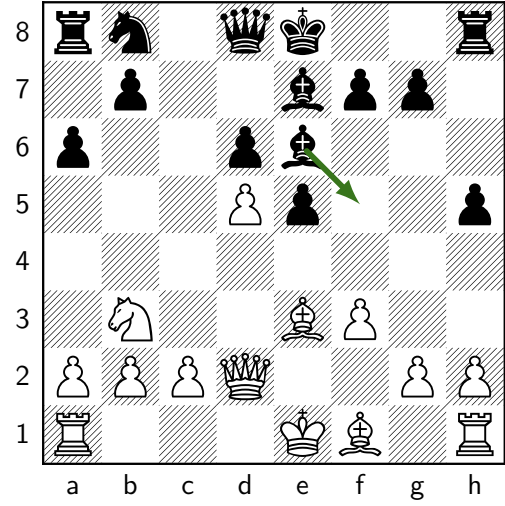


(d)

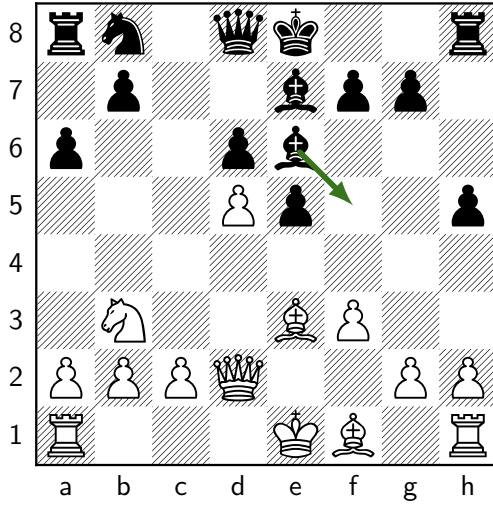
Figure 1: **Move 7, Black: Bxf6**. (a) Our position; Melanie plays Bxf6 (e7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2622` (ply 11) [structural distance  $d=35$ , QAP obj=165]: the master played Bxf6; under  $\mu$  our *Be2* plays the role of that game’s *Be2*; cross-type roles: our *nc3* as their *Ra8*, our *bf6* as their *Pd5*; transferred move e7f6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2623` (ply 11) [structural distance  $d=35$ , QAP obj=165]: the master played Bxf6; under  $\mu$  our *Be2* plays the role of that game’s *Be2*; cross-type roles: our *nc3* as their *Ra8*, our *bf6* as their *Pd5*; transferred move e7f6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1407` (ply 17) [structural distance  $d=47$ , QAP obj=156]: the master played Bxf6; under  $\mu$  our *Be2* plays the role of that game’s *Be2*; cross-type roles: our *pd4* as their *nd4*, our *qd1* as their *pf2*; transferred move e7f6 (**endorses this move**). Synthesis: 34 retrieved analogs transferred to Bxf6 (summed QAP objective 4807, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



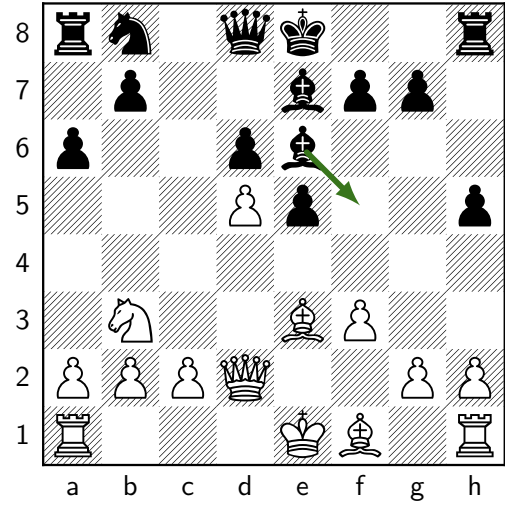
(a)



(b)

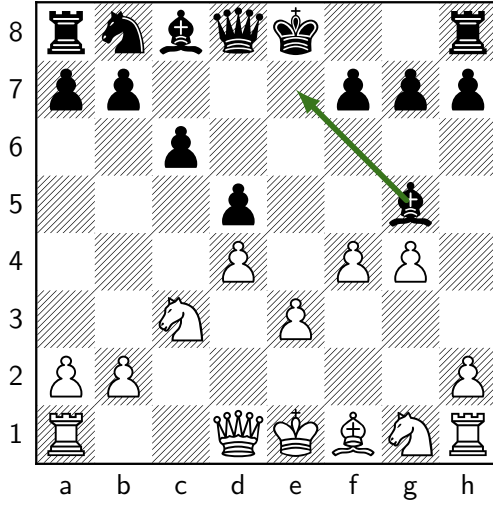


(c)

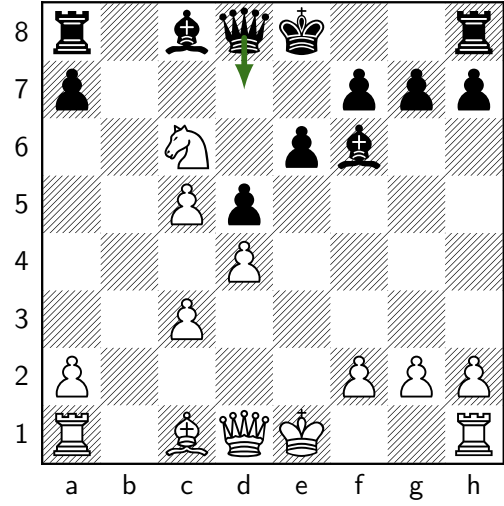


(d)

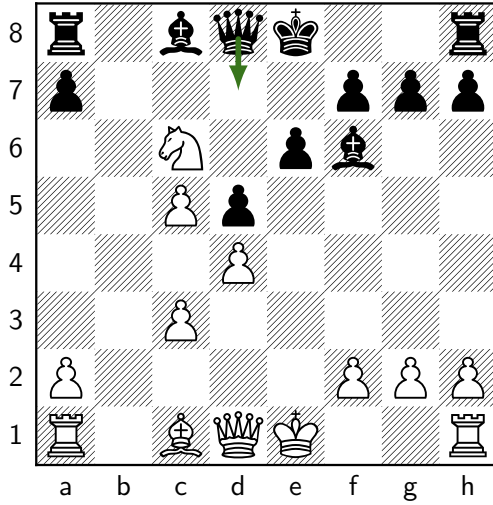
Figure 2: **Move 8, Black: Bg5**. (a) Our position; Melanie plays Bg5 (f6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0000#201 (ply 21) [structural distance  $d=55$ , QAP obj=25]: the master played Bf5; under  $\mu$  our Bf3 plays the role of that game’s Be3; cross-type roles: our Pd5 as their Nb8, our Bc8 as their Rh8; transferred move f6g5 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0000#1392 (ply 21) [structural distance  $d=55$ , QAP obj=25]: the master played Bf5; under  $\mu$  our Bf3 plays the role of that game’s Be3; cross-type roles: our Pd5 as their Nb8, our Bc8 as their Rh8; transferred move f6g5 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0001#2033 (ply 21) [structural distance  $d=55$ , QAP obj=25]: the master played Bf5; under  $\mu$  our Bf3 plays the role of that game’s Be3; cross-type roles: our Pd5 as their Nb8, our Bc8 as their Rh8; transferred move f6g5 (**endorses this move**). Synthesis: 20 retrieved analogs transferred to Bg5 (summed QAP objective 494, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



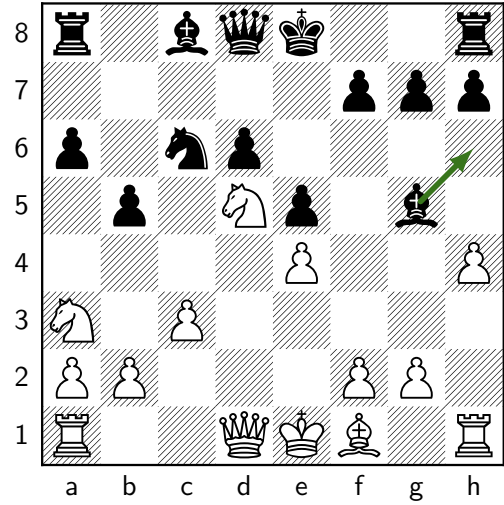
(a)



(b)

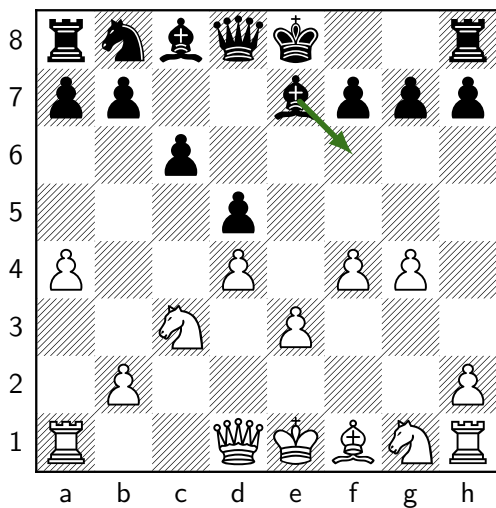


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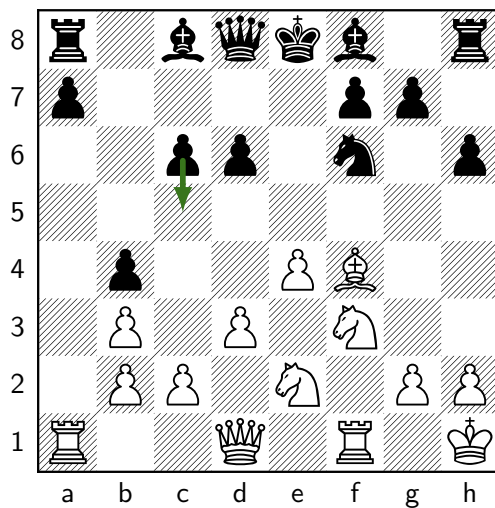


(d)

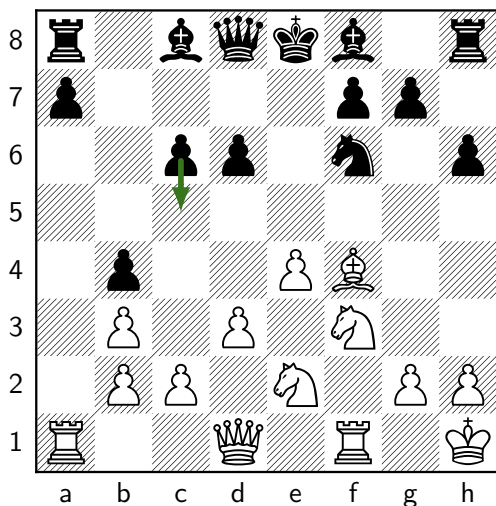
Figure 3: **Move 9, Black: Be7.** (a) Our position; Melanie plays Be7 (g5e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3405` (ply 23) [structural distance  $d=62$ , QAP obj=18]: the master played Qd7; under  $\mu$  our Bg4 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Pg7 as their Ra8, our Pd5 as their Bc8; transferred move g5e7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#3406` (ply 23) [structural distance  $d=62$ , QAP obj=18]: the master played Qd7; under  $\mu$  our Bg4 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Pg7 as their Ra8, our Pd5 as their Bc8; transferred move g5e7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2779` (ply 23) [structural distance  $d=52$ , QAP obj=38]: the master played Bh6; under  $\mu$  our Bg4 plays the role of that game’s Bg4; cross-type roles: our Nb8 as their Pf7, our pg4 as their Nc6; transferred move g5h6. Synthesis: 2 retrieved analogs transferred to Be7 (summed QAP objective 36, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



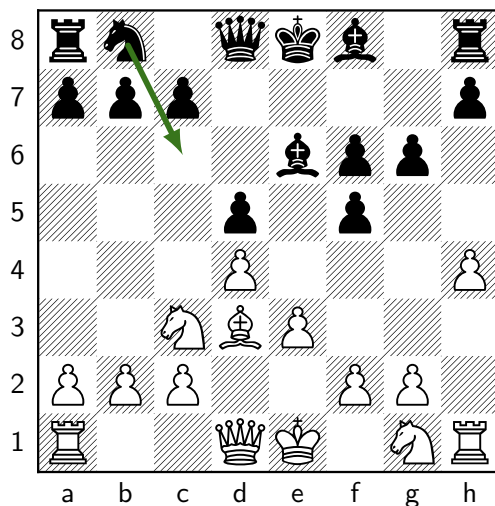
(a)



(b)

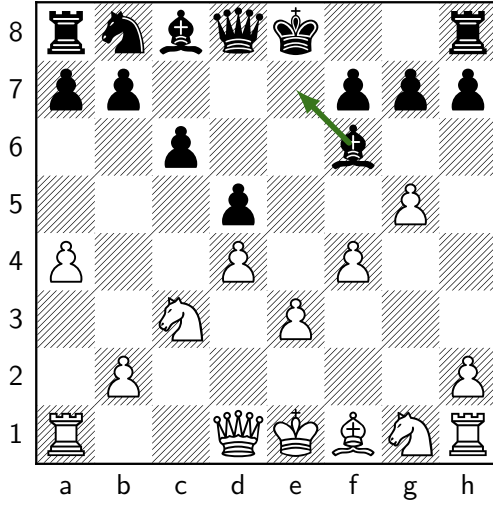


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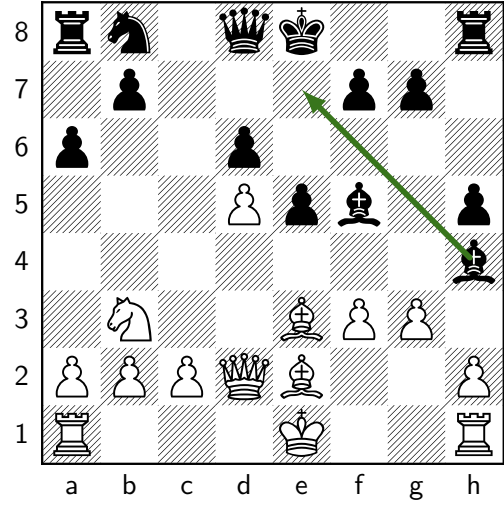


(d)

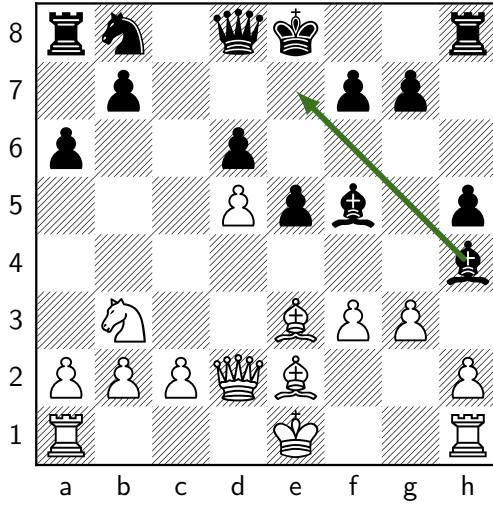
Figure 4: **Move 10, Black: Bf6**. (a) Our position; Melanie plays Bf6 (e7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0001#738 (ply 23) [structural distance  $d=50$ , QAP obj=51]: the master played c5; under  $\mu$  our *Be2* plays the role of that game’s *Pc3* (a **cross-type** correspondence); cross-type roles: our *Ph7* as their *Ra8*, our *rh1* as their *Bc8*; transferred move e7f6 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0001#739 (ply 23) [structural distance  $d=50$ , QAP obj=51]: the master played c5; under  $\mu$  our *Be2* plays the role of that game’s *Pc3* (a **cross-type** correspondence); cross-type roles: our *Ph7* as their *Ra8*, our *rh1* as their *Bc8*; transferred move e7f6 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0002#1255 (ply 13) [structural distance  $d=48$ , QAP obj=35]: the master played Nc6; under  $\mu$  our *Be2* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *qd1* as their *Rh8*, our *Ra8* as their *Pb7*; transferred move e7f6 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to Bf6 (summed QAP objective 390, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



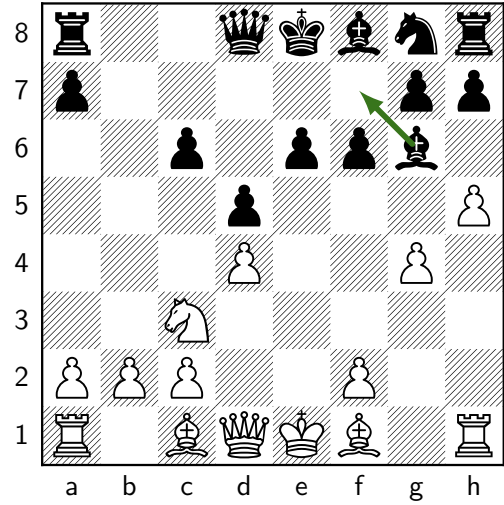
(a)



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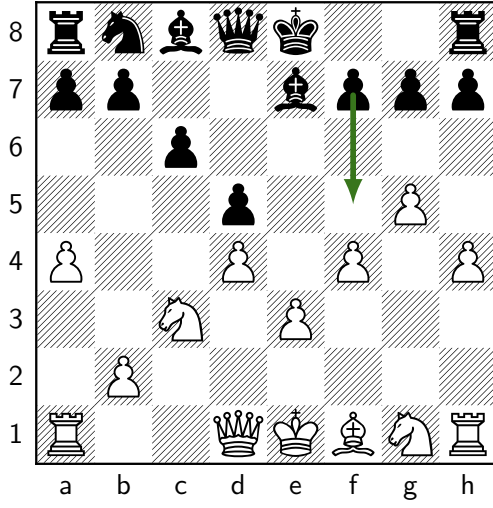


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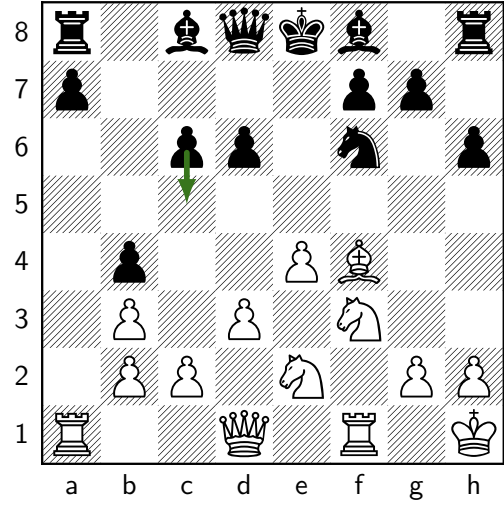


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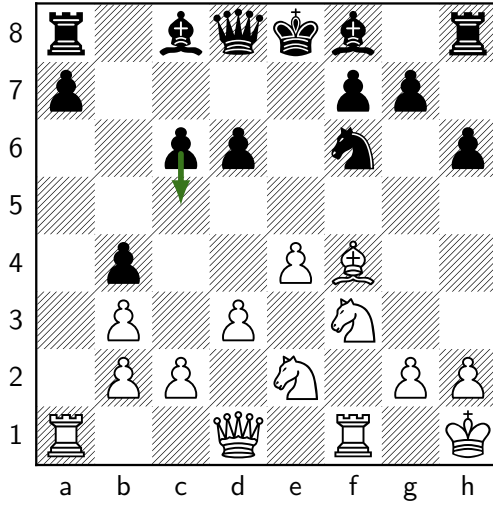
Figure 5: **Move 11, Black: Be7 (f6e7).** (a) Our position; Melanie plays Be7 (f6e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2463` (ply 25) [structural distance  $d=69$ , QAP obj=36]: the master played Be7; under  $\mu$  our Bf3 plays the role of that game’s Bh5; cross-type roles: our bf1 as their Ra8, our ph2 as their Nb8; transferred move f6e7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3555` (ply 25) [structural distance  $d=69$ , QAP obj=36]: the master played Be7; under  $\mu$  our Bf3 plays the role of that game’s Bh5; cross-type roles: our bf1 as their Ra8, our ph2 as their Nb8; transferred move f6e7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#307` (ply 21) [structural distance  $d=61$ , QAP obj=33]: the master played Bf7; under  $\mu$  our Bf3 plays the role of that game’s Bg3; cross-type roles: our ph2 as their Ra8, our Ra8 as their Bf8; transferred move f6e7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Be7 (summed QAP objective 106, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



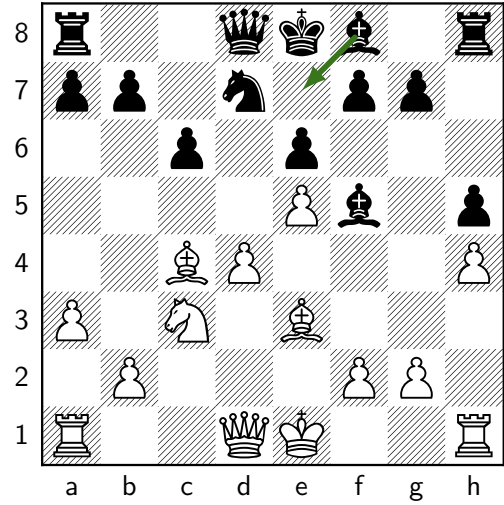
(a)



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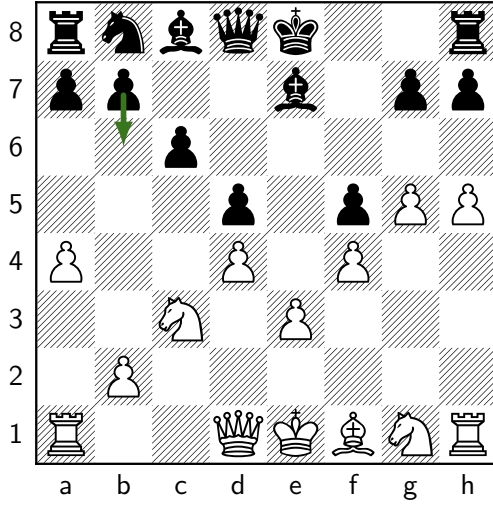


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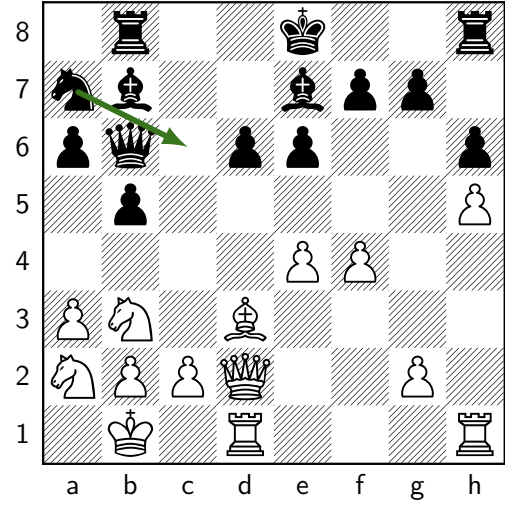


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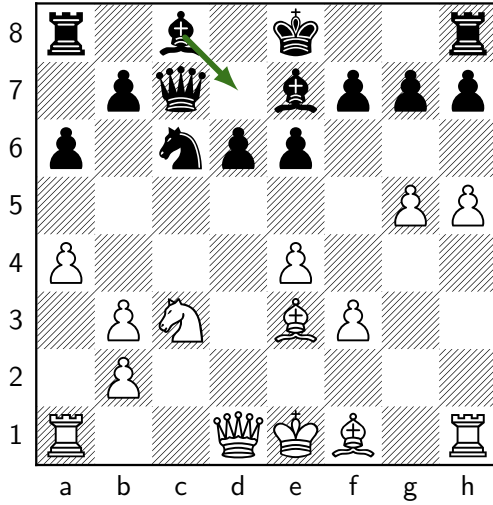
Figure 6: **Move 12, Black: f5.** (a) Our position; Melanie plays f5 (f7f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#738` (ply 23) [structural distance  $d=55$ , QAP obj=61]: the master played c5; under  $\mu$  our *Pf2* plays the role of that game’s *Pc3*; cross-type roles: our *pe3* as their *Ra8*, our *Ph7* as their *Bc8*; transferred move f7f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#739` (ply 23) [structural distance  $d=55$ , QAP obj=61]: the master played c5; under  $\mu$  our *Pf2* plays the role of that game’s *Pc3*; cross-type roles: our *pe3* as their *Ra8*, our *Ph7* as their *Bc8*; transferred move f7f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1165` (ply 23) [structural distance  $d=66$ , QAP obj=46]: the master played Be7; under  $\mu$  our *Pf2* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *Be7* as their *Qd8*, our *Bc8* as their *Rh8*; transferred move f7f5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to f5 (summed QAP objective 168, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



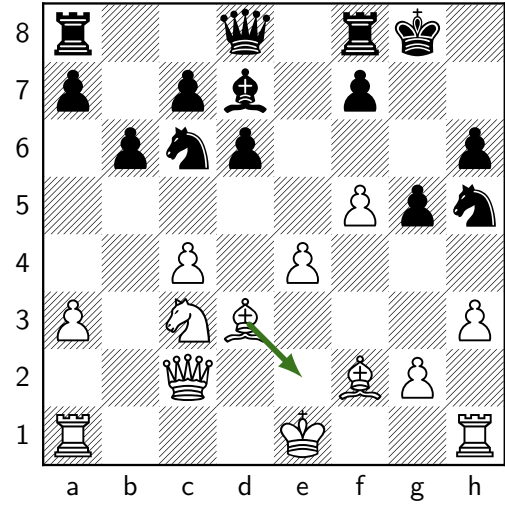
(a)



(b)



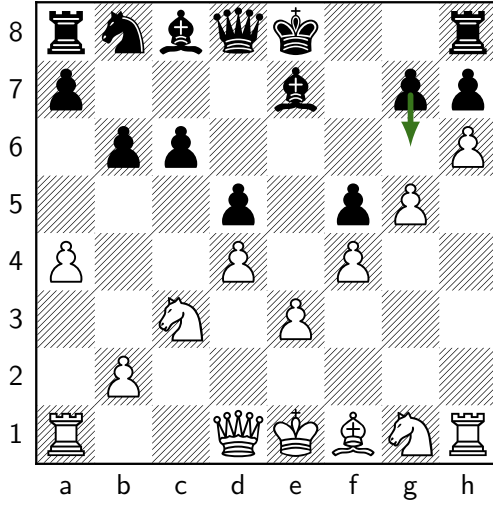
(c)



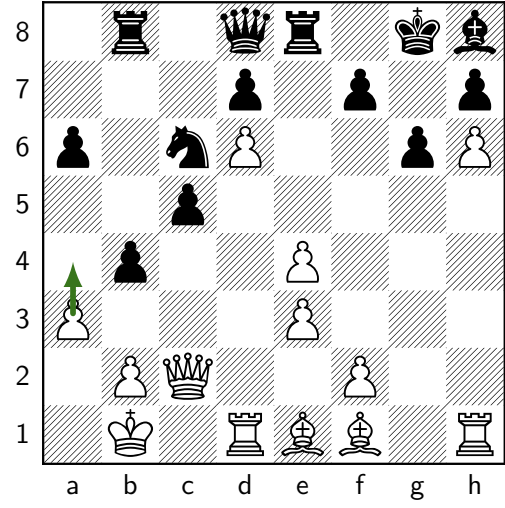
(d)

Figure 7: **Move 13, Black: b6.** (a) Our position; Melanie plays b6 (b7b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1370` (ply 37) [structural distance  $d=66$ , QAP obj=38]: the master played Nc6; under  $\mu$  our *Pb2* plays the role of that game’s *Na2* (a **cross-type** correspondence); cross-type roles: our *Qd8* as their *Rb8*, our *pd4* as their *Be7*; transferred move b7b6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1771` (ply 27) [structural distance  $d=59$ , QAP obj=37]: the master played Bd7; under  $\mu$  our *Pb2* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *ng1* as their *Ra8*, our *bf1* as their *Pb7*; transferred move b7b6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3486` (ply 34) [structural distance  $d=63$ , QAP obj=35]: the master played Be2; under  $\mu$  our *Pb2* plays the role of that game’s *Bd3* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Rh1*, our *Ra8* as their *Qc2*; transferred move b7b6 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to b6 (summed QAP objective 208, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.

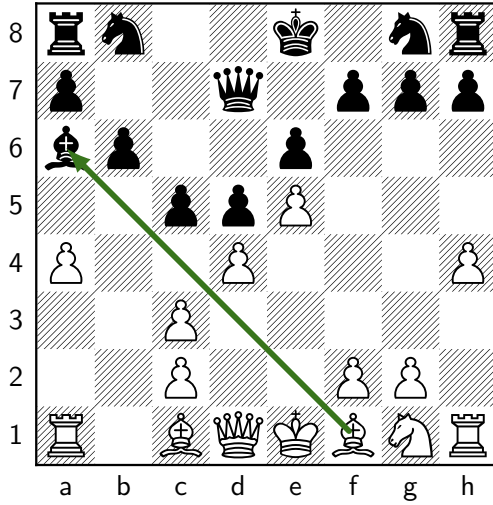




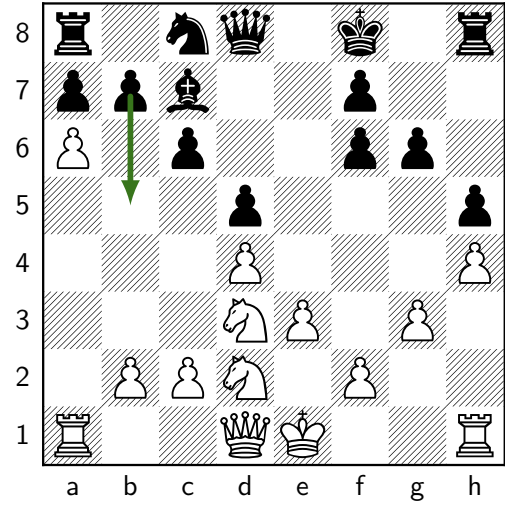
(a)



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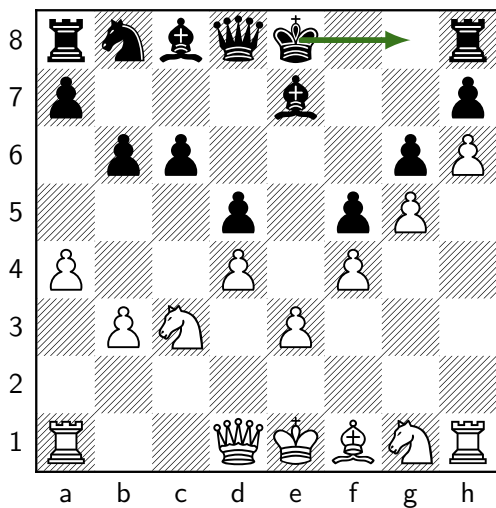


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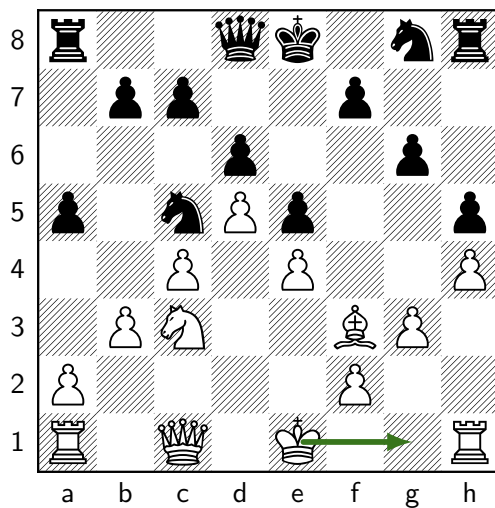


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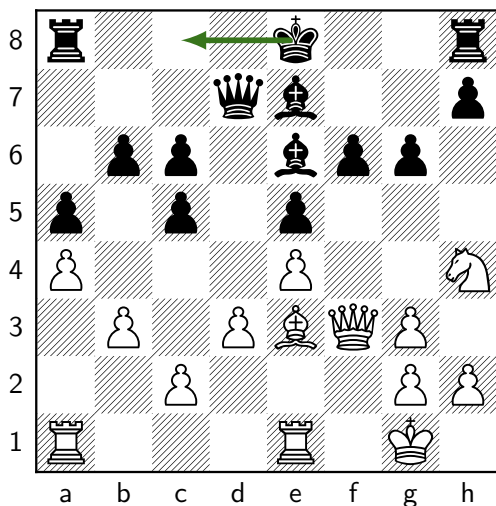
Figure 8: **Move 14, Black: g6.** (a) Our position; Melanie plays g6 (g7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1485` (ply 40) [structural distance  $d=74$ , QAP obj=60]: the master played a4; under  $\mu$  our  $Pg2$  plays the role of that game’s  $Pa3$ ; cross-type roles: our  $Qd8$  as their  $Rd1$ , our  $Pg7$  as their  $Be1$ ; transferred move g7g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#333` (ply 16) [structural distance  $d=69$ , QAP obj=58]: the master played Bxa6; under  $\mu$  our  $Pg2$  plays the role of that game’s  $Bf1$  (a **cross-type** correspondence); cross-type roles: our  $Pa7$  as their  $Ra1$ , our  $ra1$  as their  $Bc1$ ; transferred move g7g6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1354` (ply 29) [structural distance  $d=64$ , QAP obj=50]: the master played b5; under  $\mu$  our  $Pg2$  plays the role of that game’s  $Pb2$ ; cross-type roles: our  $Ph7$  as their  $Ra8$ , our  $Pb6$  as their  $Nc8$ ; transferred move g7g6 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to g6 (summed QAP objective 249, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



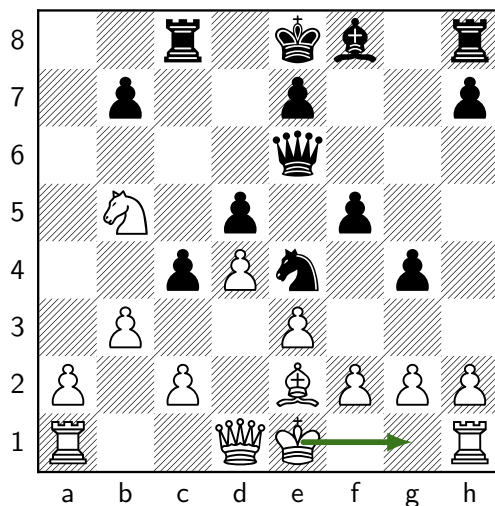
(a)



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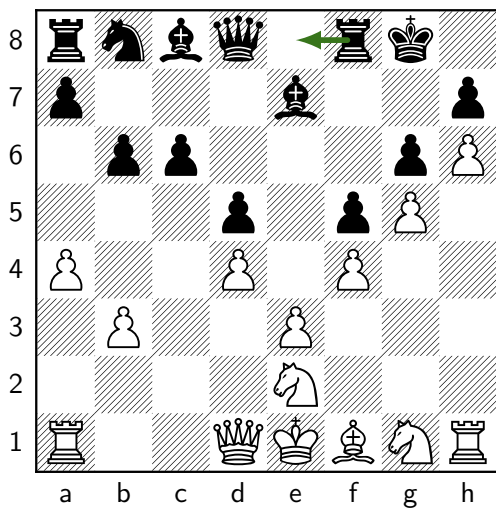


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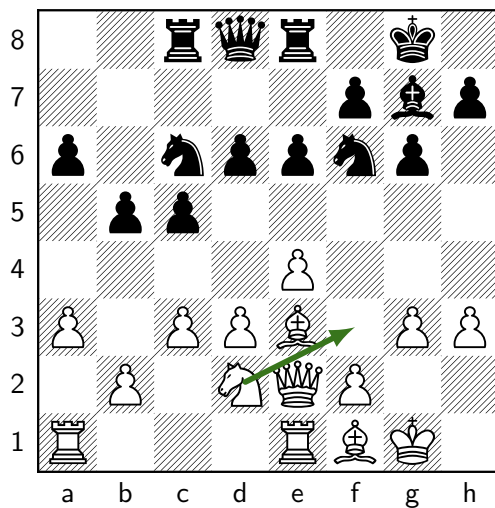


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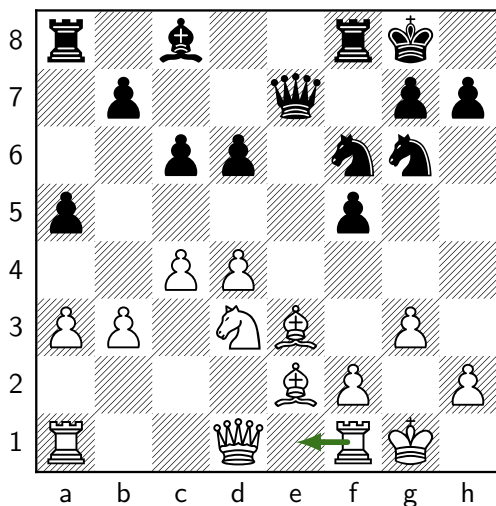
Figure 9: **Move 15, Black: 0-0.** (a) Our position; Melanie plays 0-0 (e8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#513` (ply 24) [structural distance  $d=94$ , QAP obj=61]: the master played 0-0; under  $\mu$  our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pf5* as their *Ra1*, our *Pb6* as their *Nc3*; transferred move e8g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2589` (ply 35) [structural distance  $d=86$ , QAP obj=49]: the master played 0-0-0; under  $\mu$  our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd8* as their *Rh8*, our *Be7* as their *Qd7*; transferred move e8g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3194` (ply 32) [structural distance  $d=88$ , QAP obj=48]: the master played 0-0; under  $\mu$  our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc8* as their *Ra1*, our *Be7* as their *Pb3*; transferred move e8g8 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to 0-0 (summed QAP objective 280, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



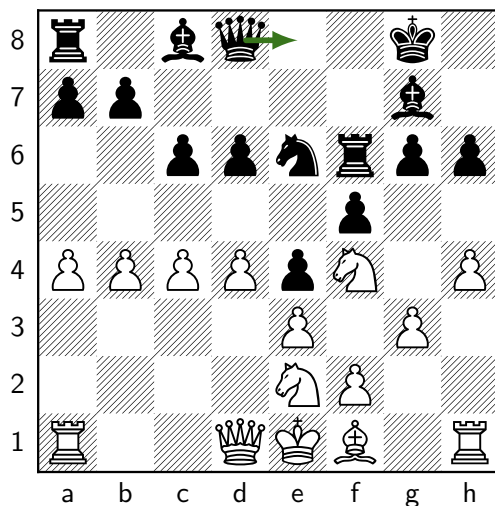
(a)



(b)

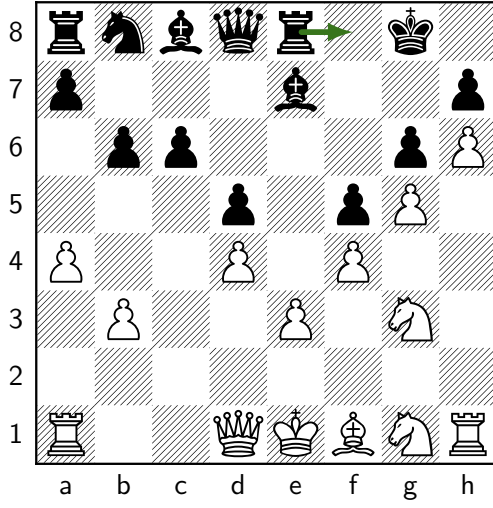


(c)

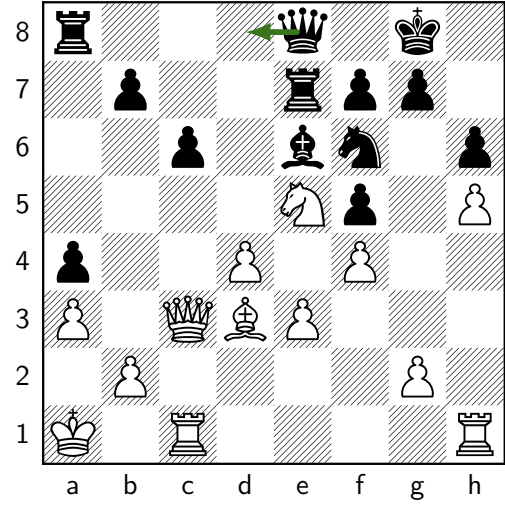


(d)

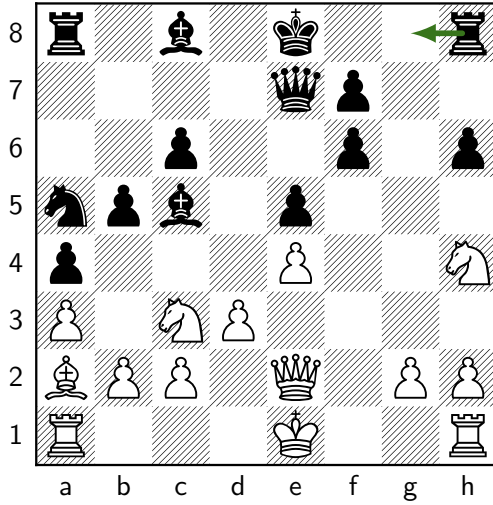
Figure 10: **Move 16, Black: Re8.** (a) Our position; Melanie plays Re8 (f8e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2398` (ply 30) [structural distance  $d=95$ , QAP obj=23]: the master played Nf3; under  $\mu$  our Rf1 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Ra1, our Pd5 as their Bf1; transferred move f8e8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#194` (ply 32) [structural distance  $d=93$ , QAP obj=18]: the master played Re1; under  $\mu$  our Rf1 plays the role of that game’s Rf1; cross-type roles: our Pb6 as their Be2, our Ra8 as their Ph2; transferred move f8e8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3629` (ply 29) [structural distance  $d=92$ , QAP obj=15]: the master played Qe8; under  $\mu$  our Rf1 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Qd8 as their Bc8, our Nb8 as their Pa7; transferred move f8e8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Re8 (summed QAP objective 70, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



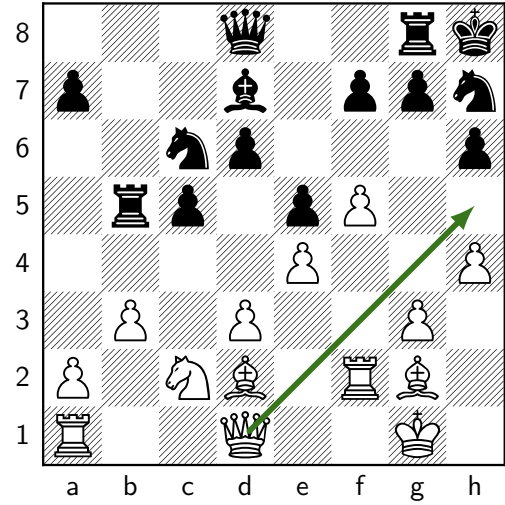
(a)



(b)

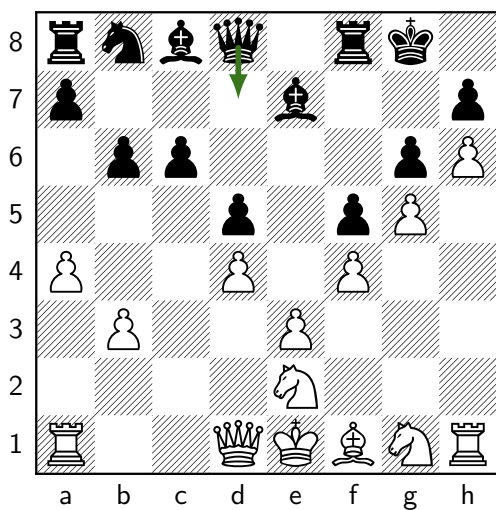


(c)

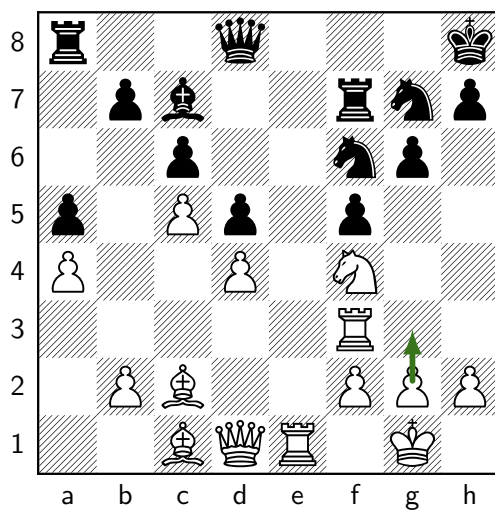


(d)

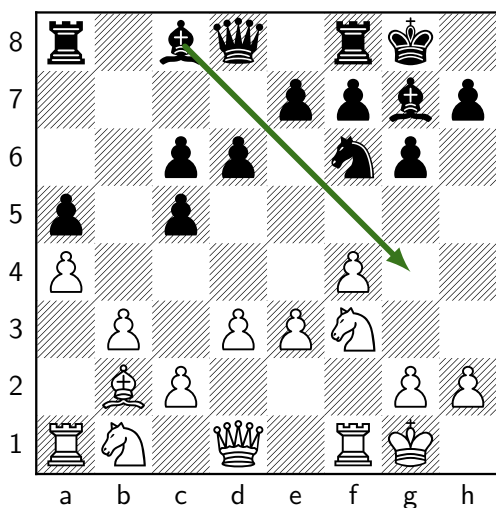
Figure 11: **Move 17, Black: Rf8**. (a) Our position; Melanie plays Rf8 (e8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0001#799 (ply 39) [structural distance  $d=87$ , QAP obj=72]: the master played Qd8; under  $\mu$  our Re1 plays the role of that game's Qe1 (a **cross-type** correspondence); cross-type roles: our Pc6 as their Ra8, our Qd8 as their Re7; transferred move e8f8 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0003#32 (ply 29) [structural distance  $d=87$ , QAP obj=57]: the master played Rg8; under  $\mu$  our Re1 plays the role of that game's Rh1; cross-type roles: our Qd8 as their Ra8, our Ph7 as their Bc8; transferred move e8f8 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0000#1324 (ply 46) [structural distance  $d=91$ , QAP obj=42]: the master played Qh5; under  $\mu$  our Re1 plays the role of that game's Qd1 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Ra1, our Qd8 as their Nc2; transferred move e8f8 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Rf8 (summed QAP objective 295, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



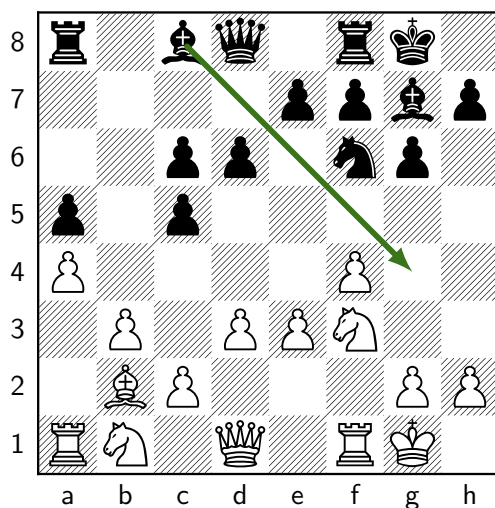
(a)



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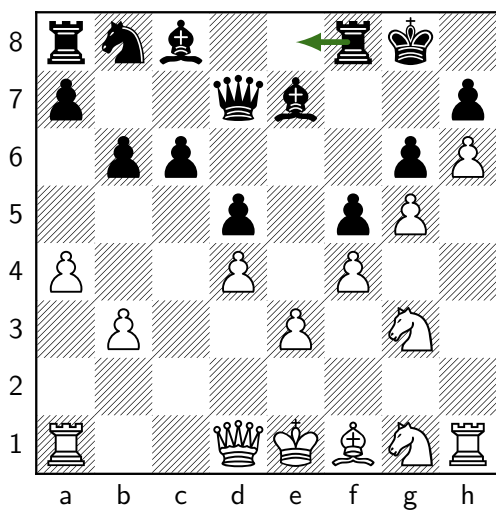


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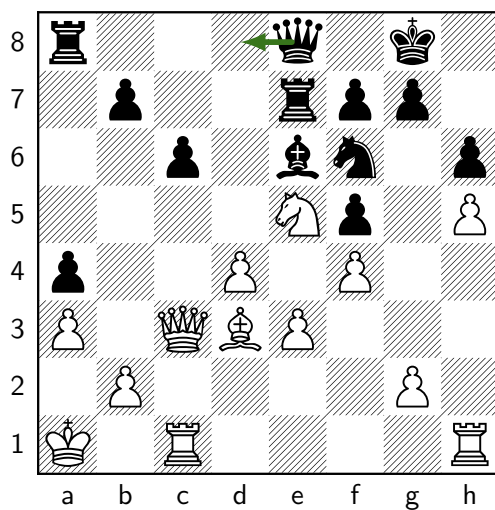


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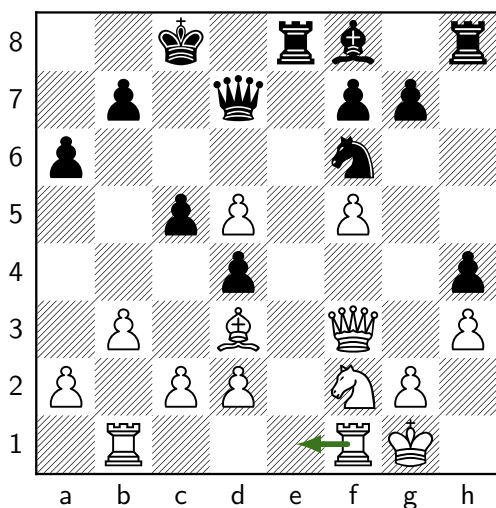
Figure 12: **Move 18, Black: Qd7.** (a) Our position; Melanie plays Qd7 (d8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2272` (ply 48) [structural distance  $d=95$ , QAP obj=18]: the master played g3; under  $\mu$  our Qd1 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our Pb6 as their Bc1, our Pf5 as their Qd1; transferred move d8d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1402` (ply 19) [structural distance  $d=90$ , QAP obj=16]: the master played Bg4; under  $\mu$  our Qd1 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Qd8, our Pd5 as their Rf8; transferred move d8d7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1403` (ply 19) [structural distance  $d=90$ , QAP obj=16]: the master played Bg4; under  $\mu$  our Qd1 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Qd8, our Pd5 as their Rf8; transferred move d8d7 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Qd7 (summed QAP objective 109, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



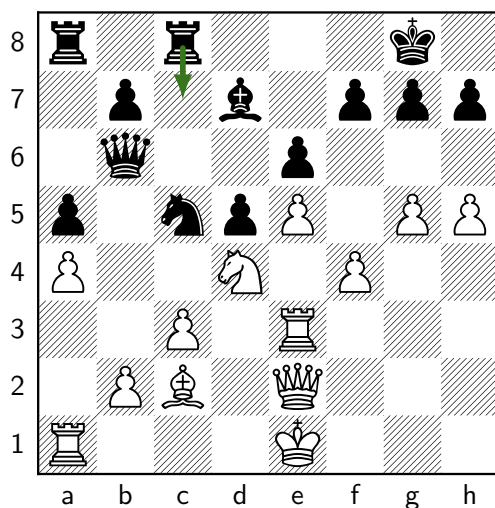
(a)



(b)

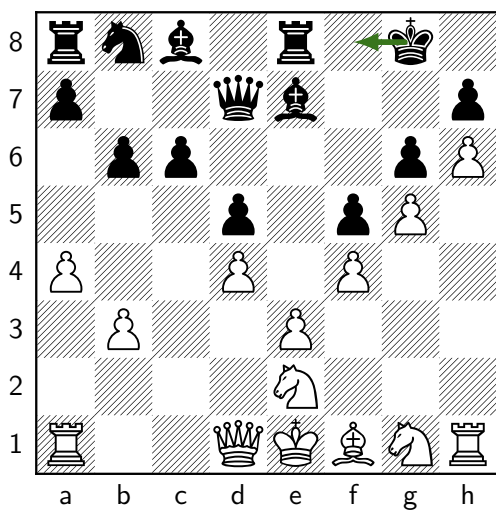


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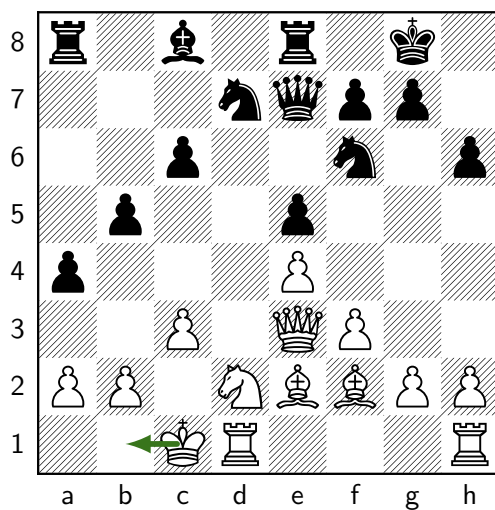


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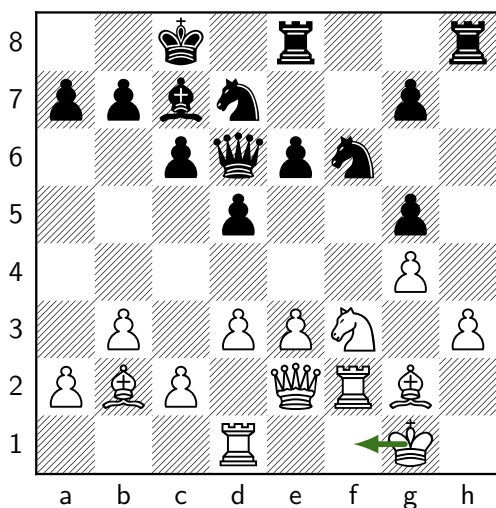
Figure 13: **Move 19, Black: Re8**. (a) Our position; Melanie plays Re8 (f8e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0001#799 (ply 39) [structural distance  $d=85$ , QAP obj=73]: the master played Qd8; under  $\mu$  our Rf1 plays the role of that game’s Qe1 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Ra8, our Qd7 as their Pb7; transferred move f8e8 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0003#185 (ply 36) [structural distance  $d=86$ , QAP obj=67]: the master played Rfe1; under  $\mu$  our Rf1 plays the role of that game’s Rf1; cross-type roles: our Nb8 as their Rb1, our Ra8 as their Pa2; transferred move f8e8 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0001#3420 (ply 53) [structural distance  $d=87$ , QAP obj=54]: the master played Rc7; under  $\mu$  our Rf1 plays the role of that game’s Rc1; cross-type roles: our ph6 as their Ra8, our Be7 as their Nc5; transferred move f8e8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Re8 (summed QAP objective 238, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



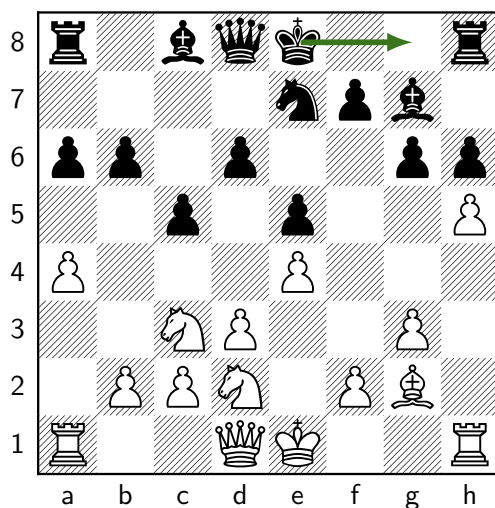
(a)



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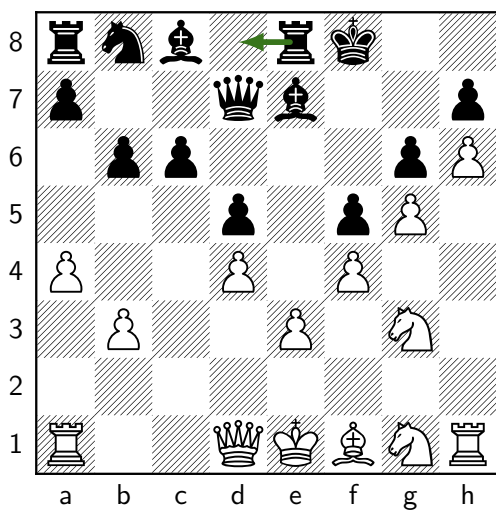
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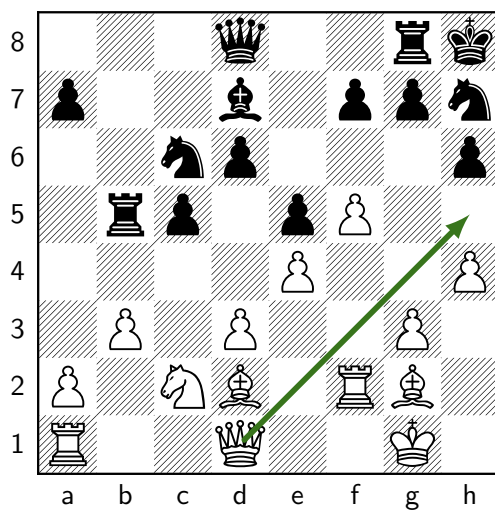
(d)

Figure 14: **Move 20, Black: Kf8.** (a) Our position; Melanie plays Kf8 (g8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#990` (ply 36) [structural distance  $d=114$ , QAP obj=24]: the master played Kb1; under  $\mu$  our Kg1 plays the role of that game’s Kc1; cross-type roles: our Bc8 as their Rh1, our Pg6 as their Nd2; transferred move g8f8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3354` (ply 38) [structural distance  $d=119$ , QAP obj=20]: the master played Kf1; under  $\mu$  our Kg1 plays the role of that game’s Kg1; cross-type roles: our ng1 as their Pc2, our pa4 as their Qe2; transferred move g8f8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#393` (ply 23) [structural distance  $d=114$ , QAP obj=13]: the master played 0-0; under  $\mu$  our Kg1 plays the role of that game’s Ke1; cross-type roles: our ne2 as their Bc8, our pf4 as their Qd8; transferred move g8f8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kf8 (summed QAP objective 58, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

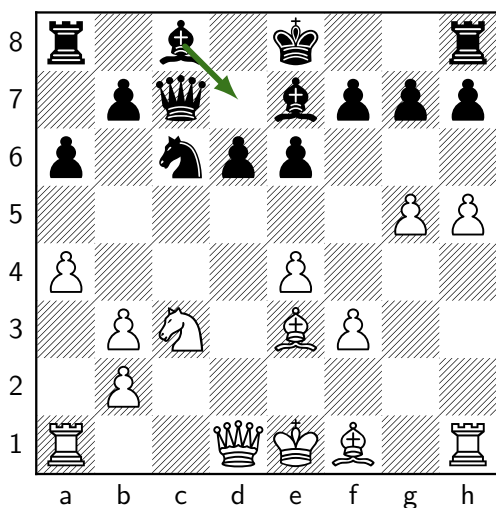




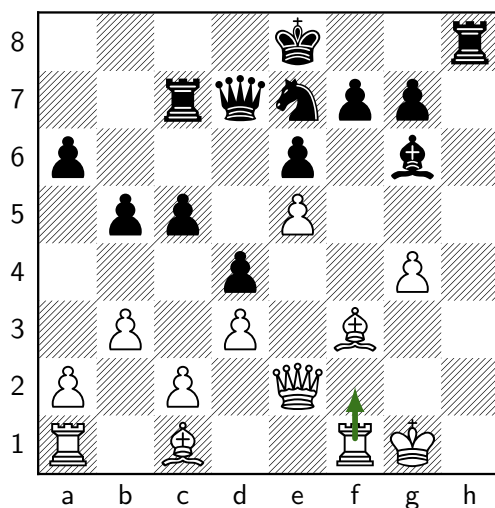
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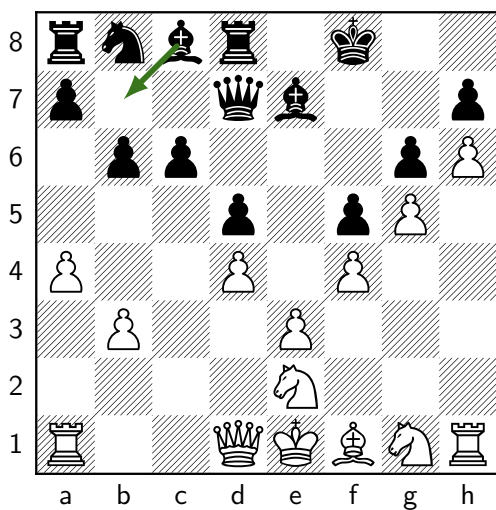
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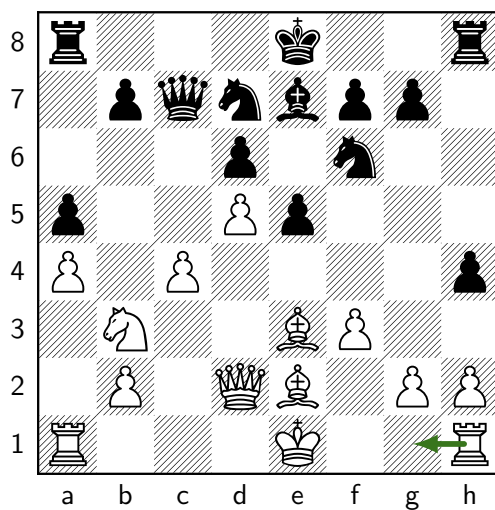
(d)

Figure 15: **Move 21, Black: Rd8.** (a) Our position; Melanie plays Rd8 (e8d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1324` (ply 46) [structural distance  $d=102$ , QAP obj=42]: the master played Qh5; under  $\mu$  our *Re1* plays the role of that game's *Qd1* (a **cross-type** correspondence); cross-type roles: our *ph6* as their *Ra1*, our *pb3* as their *Nc2*; transferred move e8d8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1771` (ply 27) [structural distance  $d=93$ , QAP obj=35]: the master played Bd7; under  $\mu$  our *Re1* plays the role of that game's *Bc1* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Ra8*, our *Pa7* as their *Rh8*; transferred move e8d8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#441` (ply 46) [structural distance  $d=101$ , QAP obj=27]: the master played Rf2; under  $\mu$  our *Re1* plays the role of that game's *Rf1*; cross-type roles: our *ng3* as their *bg6*, our *bf1* as their *rc7*; transferred move e8d8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Rd8 (summed QAP objective 149, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

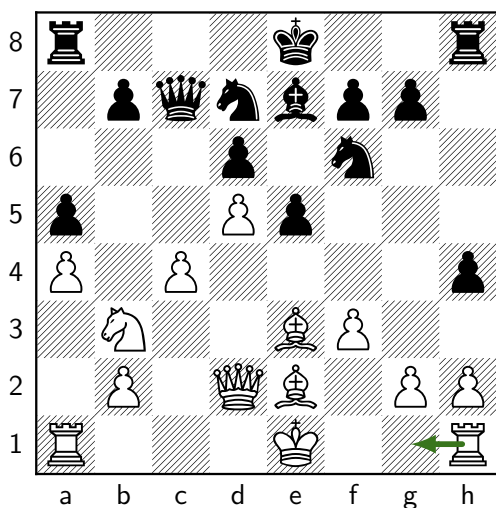




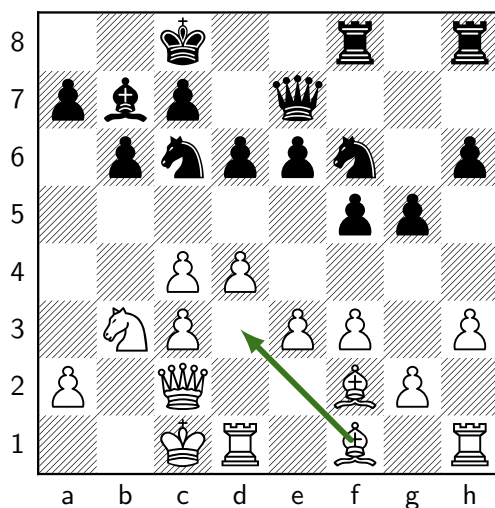
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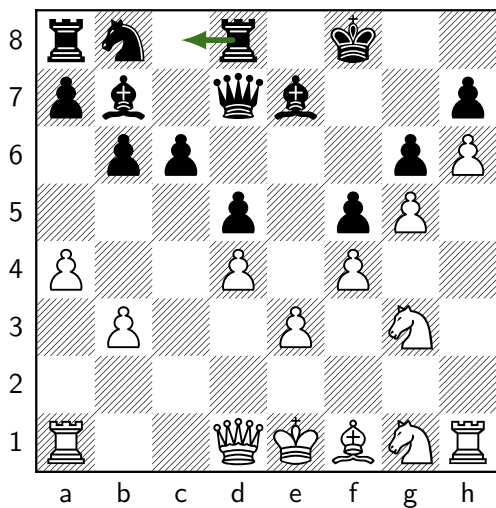


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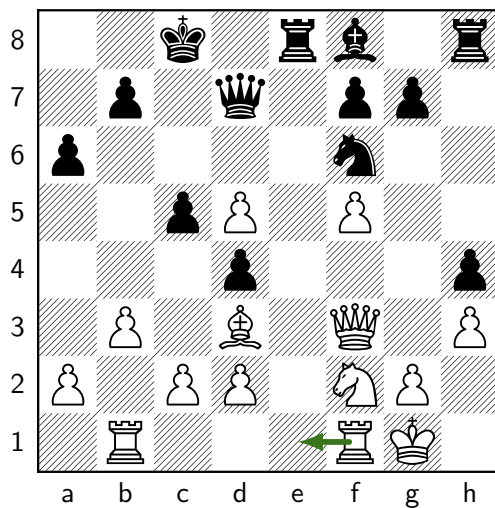


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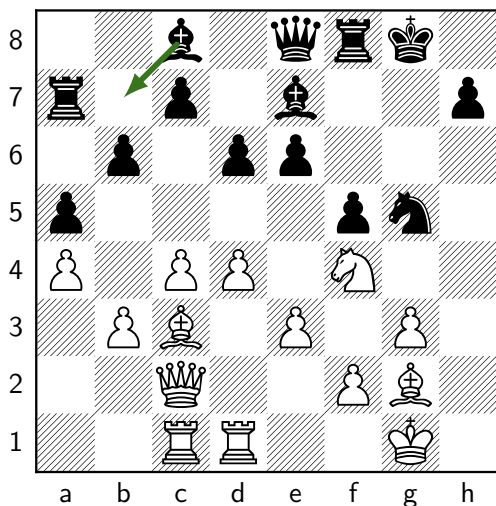
Figure 16: **Move 22, Black: Bb7**. (a) Our position; Melanie plays Bb7 (c8b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2193` (ply 28) [structural distance  $d=109$ , QAP obj=45]: the master played Rg1; under  $\mu$  our Bc1 plays the role of that game’s Rh1 (a **cross-type** correspondence); cross-type roles: our Ra8 as their Pb2, our Be7 as their Qd2; transferred move c8b7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2883` (ply 28) [structural distance  $d=109$ , QAP obj=45]: the master played Rg1; under  $\mu$  our Bc1 plays the role of that game’s Rh1 (a **cross-type** correspondence); cross-type roles: our Ra8 as their Pb2, our Be7 as their Qd2; transferred move c8b7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#1955` (ply 32) [structural distance  $d=111$ , QAP obj=28]: the master played Bd3; under  $\mu$  our Bc1 plays the role of that game’s Bf1; cross-type roles: our Pg6 as their Rd1, our ph6 as their Rh1; transferred move c8b7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bb7 (summed QAP objective 131, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



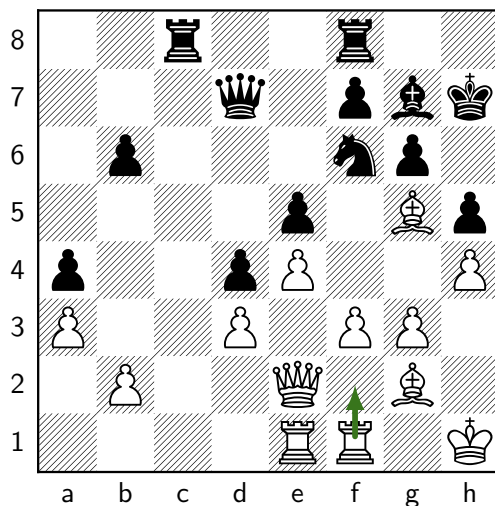
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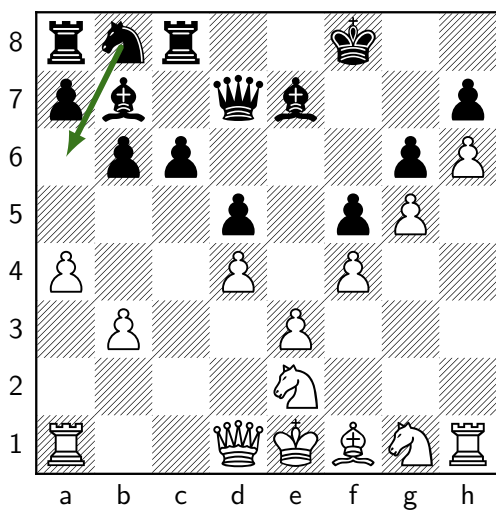


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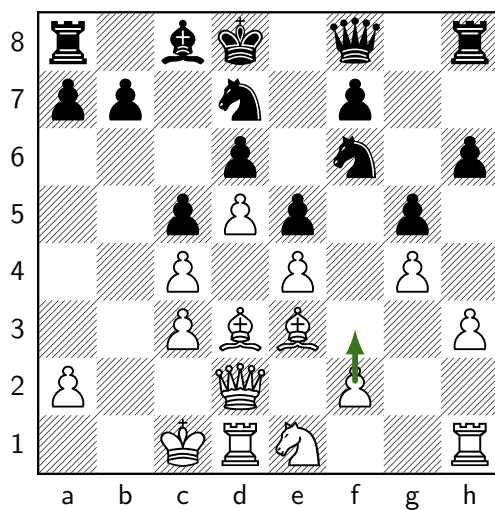


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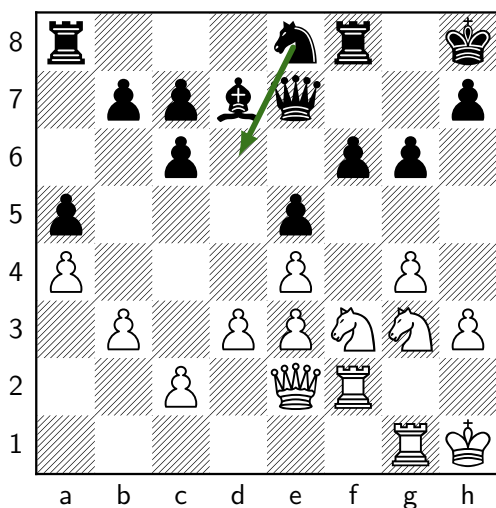
Figure 17: **Move 23, Black: Rc8.** (a) Our position; Melanie plays Rc8 (d8c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#185` (ply 36) [structural distance  $d=91$ , QAP obj=65]: the master played Rfe1; under  $\mu$  our Rd1 plays the role of that game's Rf1; cross-type roles: our Nb8 as their Rb1, our pb3 as their Nf2; transferred move d8c8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2804` (ply 47) [structural distance  $d=91$ , QAP obj=44]: the master played Bb7; under  $\mu$  our Rd1 plays the role of that game's Bc1 (a **cross-type** correspondence); cross-type roles: our Be7 as their Qe8, our ph6 as their Rf8; transferred move d8c8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#907` (ply 58) [structural distance  $d=89$ , QAP obj=10]: the master played Rf2; under  $\mu$  our Rd1 plays the role of that game's Rf1; cross-type roles: our Pa7 as their Re1, our rh1 as their Qe2; transferred move d8c8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rc8 (summed QAP objective 119, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



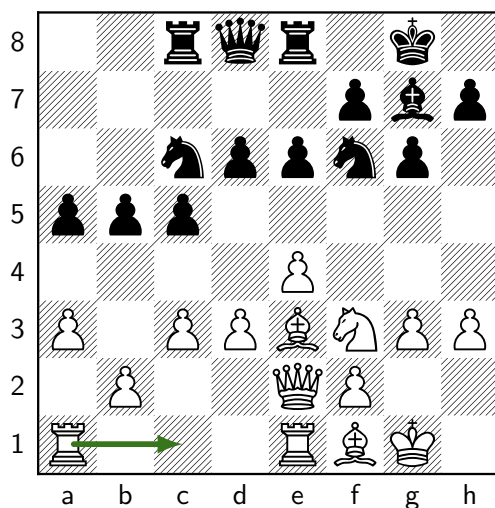
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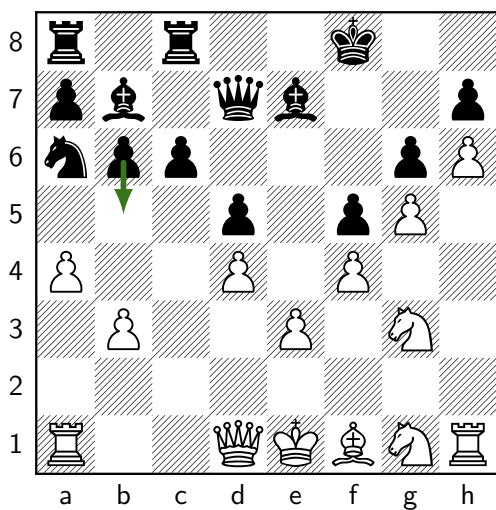


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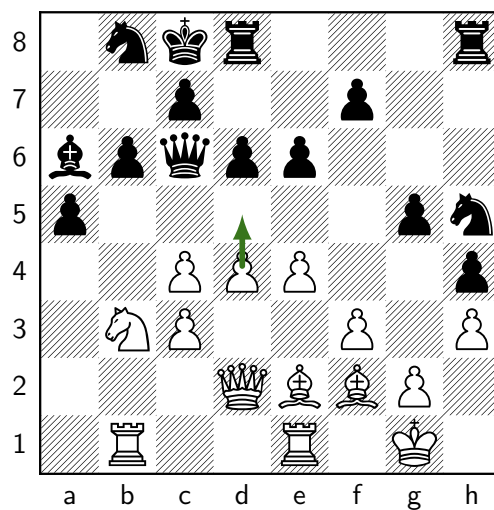


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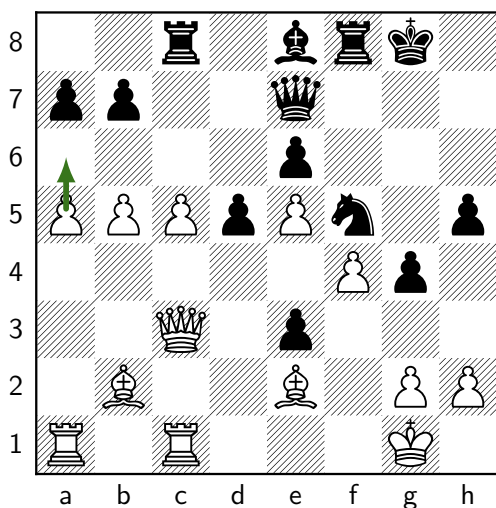
Figure 18: **Move 24, Black: Na6**. (a) Our position; Melanie plays Na6 (b8a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1351` (ply 30) [structural distance  $d=106$ , QAP obj=54]: the master played f3; under  $\mu$  our Nb1 plays the role of that game’s Pf2 (a **cross-type** correspondence); cross-type roles: our Pd5 as their Ne1, our ph6 as their Qd2; transferred move b8a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1468` (ply 45) [structural distance  $d=108$ , QAP obj=28]: the master played Nd6; under  $\mu$  our Nb1 plays the role of that game’s Ne1; cross-type roles: our ne2 as their Ra8, our Qd7 as their Rf8; transferred move b8a6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2398` (ply 32) [structural distance  $d=84$ , QAP obj=24]: the master played Rac1; under  $\mu$  our Nb1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Pf5 as their Bf1, our bf1 as their Pb2; transferred move b8a6 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Na6 (summed QAP objective 143, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



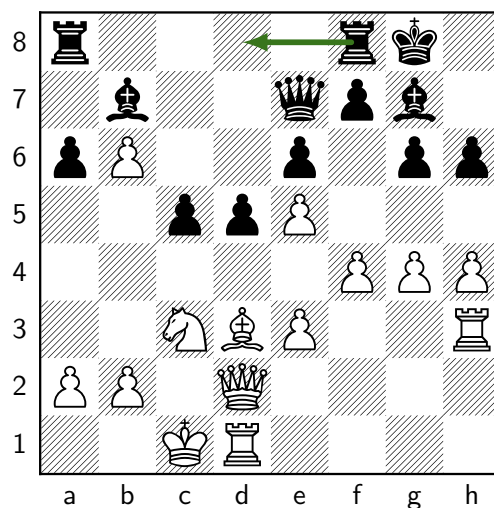
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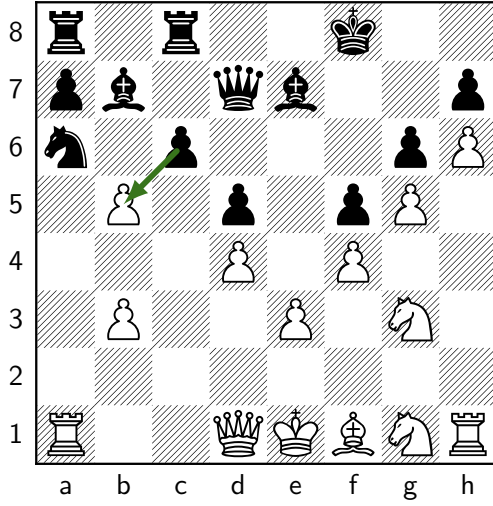


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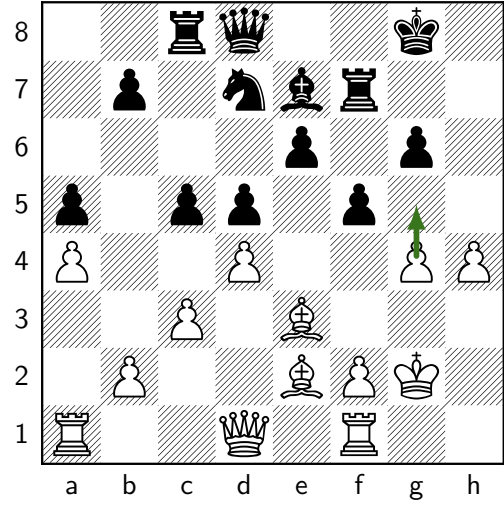


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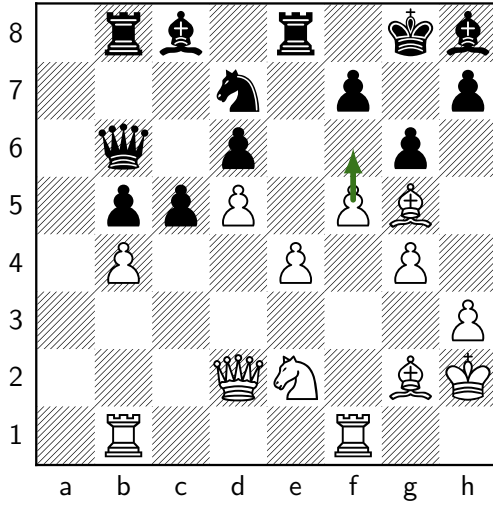
Figure 19: **Move 25, Black: b5**. (a) Our position; Melanie plays b5 (b6b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2792` (ply 40) [structural distance  $d=93$ , QAP obj=78]: the master played d5; under  $\mu$  our  $Pb3$  plays the role of that game’s  $Pd4$ ; cross-type roles: our  $Pc6$  as their  $Rb1$ , our  $bf1$  as their  $Qd2$ ; transferred move b6b5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1005` (ply 52) [structural distance  $d=101$ , QAP obj=59]: the master played a6; under  $\mu$  our  $Pb3$  plays the role of that game’s  $Pa5$ ; cross-type roles: our  $qd1$  as their  $Ra1$ , our  $Qd7$  as their  $Rc1$ ; transferred move b6b5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#717` (ply 35) [structural distance  $d=96$ , QAP obj=55]: the master played Rfd8; under  $\mu$  our  $Rc1$  plays the role of that game’s  $Rf1$ ; cross-type roles: our  $Qd7$  as their  $Ra8$ , our  $Pg6$  as their  $Bb7$ ; transferred move c8c7. Synthesis: 2 retrieved analogs transferred to b5 (summed QAP objective 137, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



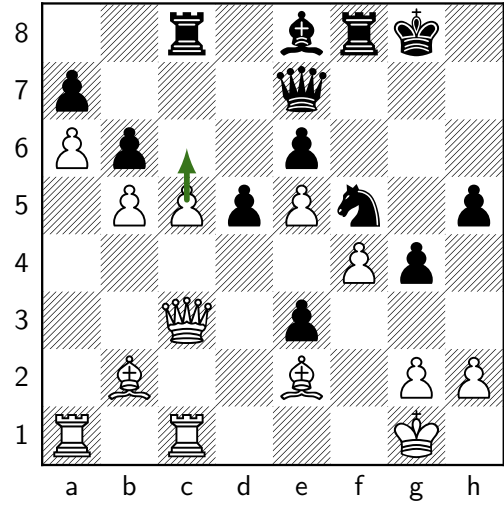
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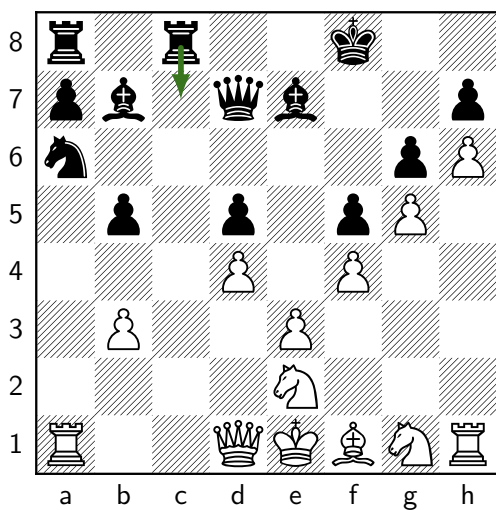


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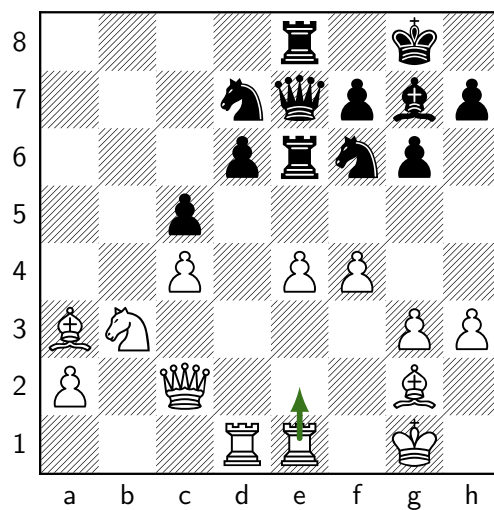


(d)

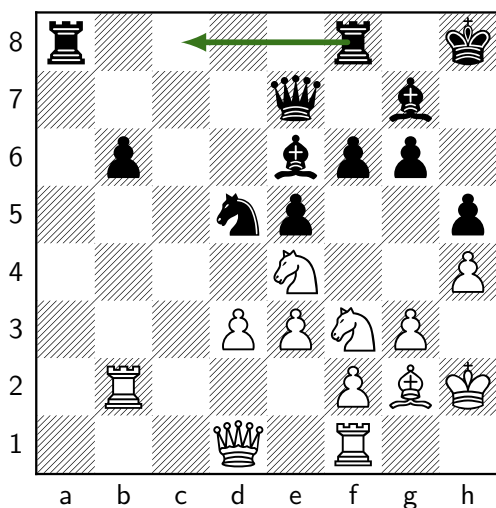
Figure 20: **Move 26, Black: cxb5.** (a) Our position; Melanie plays **cxb5** (**c6b5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#748` (ply 42) [structural distance  $d=91$ , QAP obj=29]: the master played **g5**; under  $\mu$  our *Pc3* plays the role of that game’s *Pg4*; cross-type roles: our *pg5* as their *Ra1*, our *Ph7* as their *Rf1*; transferred move **c6b5** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3679` (ply 58) [structural distance  $d=85$ , QAP obj=24]: the master played **f6**; under  $\mu$  our *Pc3* plays the role of that game’s *Pf5*; cross-type roles: our *pd4* as their *Rb1*, our *Ph7* as their *Rf1*; transferred move **c6b5** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1005` (ply 54) [structural distance  $d=82$ , QAP obj=22]: the master played **c6**; under  $\mu$  our *Pc3* plays the role of that game’s *Pc5*; cross-type roles: our *Bb7* as their *Ra1*, our *Be7* as their *Rc1*; transferred move **c6b5** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **cxb5** (summed QAP objective 75, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



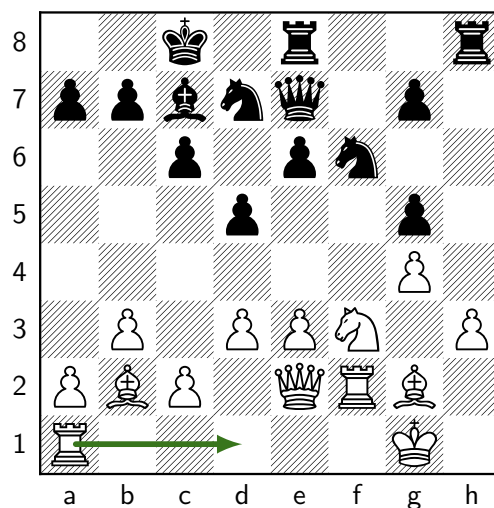
(a)



(b)

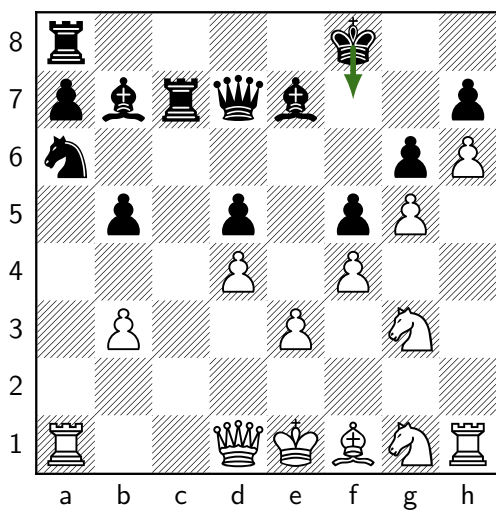


(c)

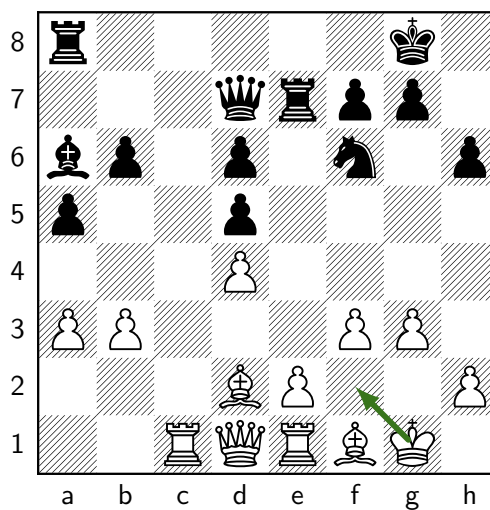


(d)

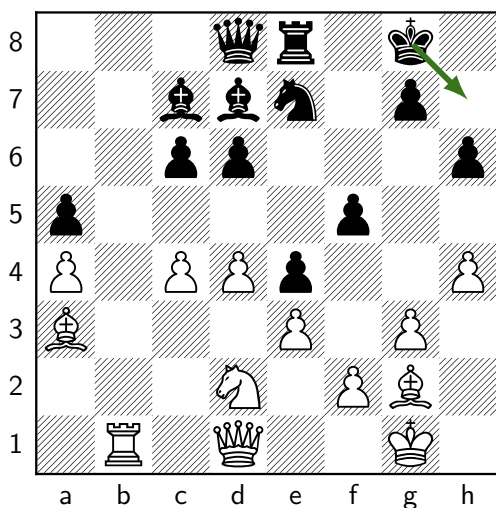
Figure 21: **Move 27, Black: Rc7**. (a) Our position; Melanie plays Rc7 (c8c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1067` (ply 58) [structural distance  $d=127$ , QAP obj=61]: the master played Re2; under  $\mu$  our Rc1 plays the role of that game's Re1; cross-type roles: our Bb7 as their Rd1, our Qd7 as their Pa2; transferred move c8c7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2625` (ply 41) [structural distance  $d=129$ , QAP obj=44]: the master played Rfc8; under  $\mu$  our Rc1 plays the role of that game's Rf1; cross-type roles: our pf4 as their Ra8, our Bb7 as their Qe7; transferred move c8c7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3354` (ply 36) [structural distance  $d=111$ , QAP obj=33]: the master played Rd1; under  $\mu$  our Rc1 plays the role of that game's Ra1; cross-type roles: our ra1 as their Bb2, our Pg6 as their Rf2; transferred move c8c7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rc7 (summed QAP objective 137, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



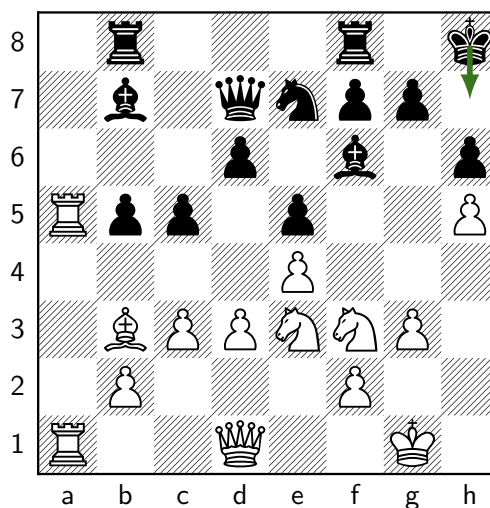
(a)



(b)



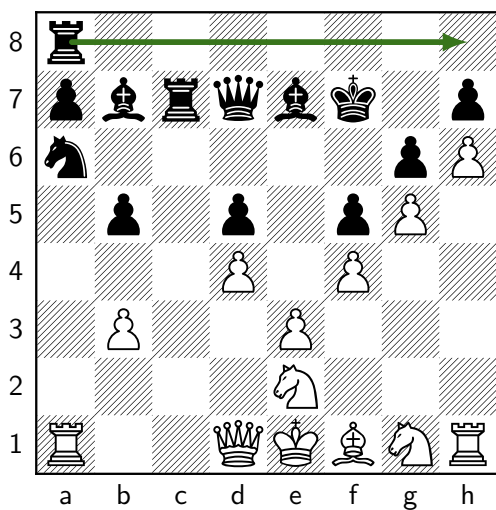
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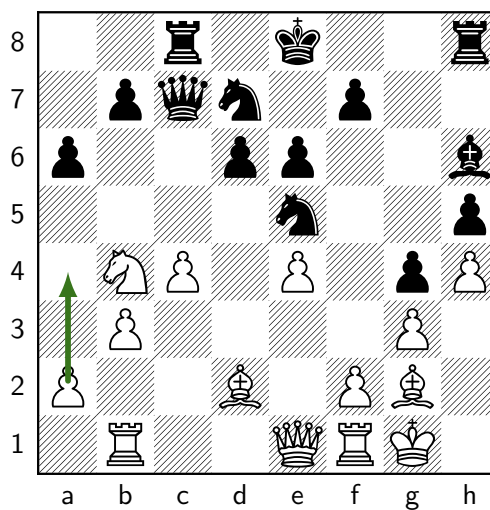
(d)

Figure 22: **Move 28, Black: Kf7.** (a) Our position; Melanie plays Kf7 (f8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2610` (ply 50) [structural distance  $d=83$ , QAP obj=71]: the master played Kf2; under  $\mu$  our Kf1 plays the role of that game's Kg1; cross-type roles: our Bb7 as their Re1, our ph6 as their Bf1; transferred move f8f7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1463` (ply 57) [structural distance  $d=82$ , QAP obj=56]: the master played Kh7; under  $\mu$  our Kf1 plays the role of that game's Kg1; cross-type roles: our Rc7 as their Qd8, our Be7 as their Re8; transferred move f8f7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1018` (ply 69) [structural distance  $d=81$ , QAP obj=44]: the master played Kh7; under  $\mu$  our Kf1 plays the role of that game's Kh1; cross-type roles: our Be7 as their Rf8, our pd4 as their Qd7; transferred move f8f7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kf7 (summed QAP objective 242, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

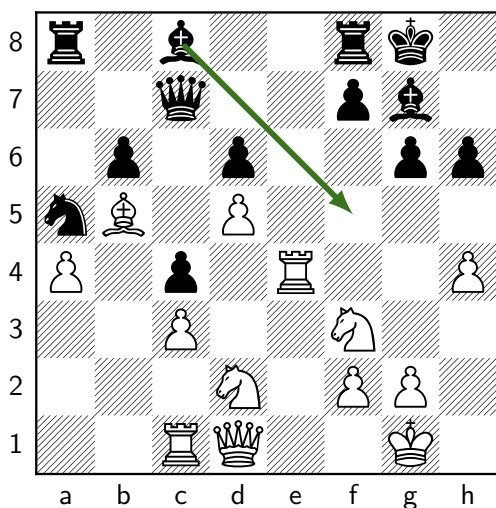




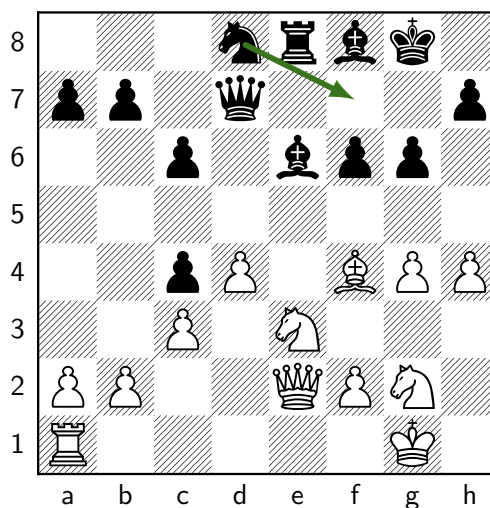
(a)



(b)

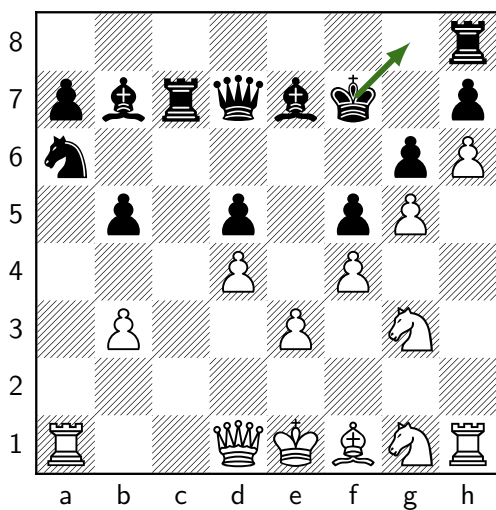


(c)

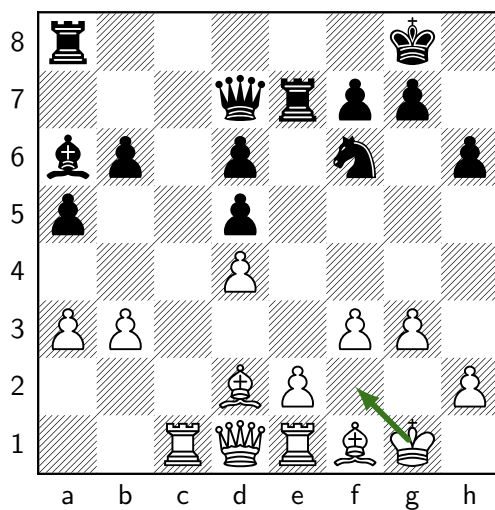


(d)

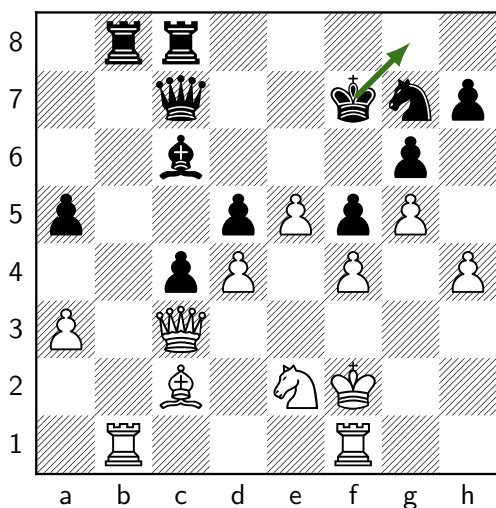
Figure 23: **Move 29, Black: Rh8.** (a) Our position; Melanie plays Rh8 (a8h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#178` (ply 40) [structural distance  $d=109$ , QAP obj=37]: the master played a4; under  $\mu$  our *Ra1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *ph6* as their *Rb1*, our *Ph7* as their *Qe1*; transferred move a8h8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#773` (ply 41) [structural distance  $d=104$ , QAP obj=32]: the master played Bf5; under  $\mu$  our *Ra1* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *Pf5* as their *Ra8*, our *Pb5* as their *Rf8*; transferred move a8h8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1065` (ply 51) [structural distance  $d=107$ , QAP obj=48]: the master played Nf7; under  $\mu$  our *Qd2* plays the role of that game’s *Nd1* (a **cross-type** correspondence); cross-type roles: our *Pg6* as their *Bf8*, our *Ra8* as their *Pa7*; transferred move d7e8. Synthesis: 2 retrieved analogs transferred to Rh8 (summed QAP objective 69, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



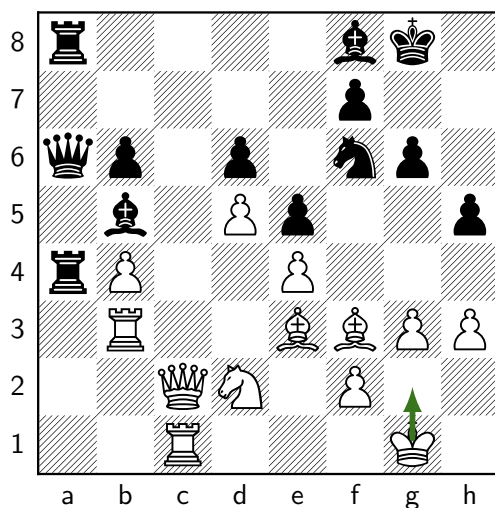
(a)



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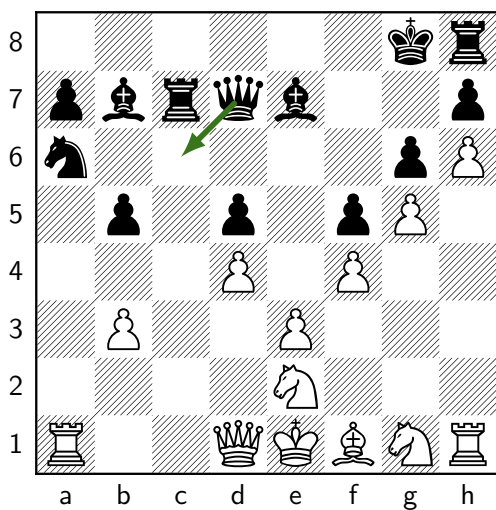


(c)

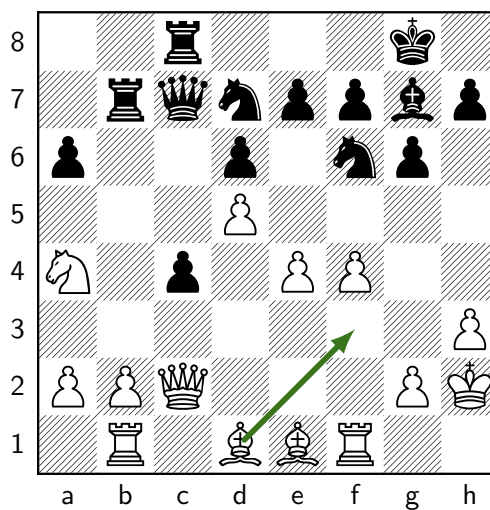


(d)

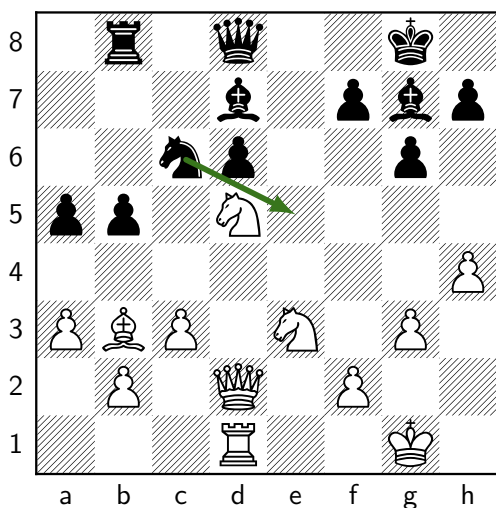
Figure 24: **Move 30, Black: Kg8.** (a) Our position; Melanie plays Kg8 (f7g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2610` (ply 50) [structural distance  $d=89$ , QAP obj=70]: the master played Kf2; under  $\mu$  our Kf2 plays the role of that game’s Kg1; cross-type roles: our qd1 as their Rc1, our pg5 as their Re1; transferred move f7g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#329` (ply 67) [structural distance  $d=86$ , QAP obj=46]: the master played Kg8; under  $\mu$  our Kf2 plays the role of that game’s Kf2; cross-type roles: our Na6 as their Rb8, our Qd7 as their Rc8; transferred move f7g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2720` (ply 54) [structural distance  $d=88$ , QAP obj=42]: the master played Kg2; under  $\mu$  our Kf2 plays the role of that game’s Kg1; cross-type roles: our qd1 as their Nd2, our ph6 as their Rb3; transferred move f7g8 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kg8 (summed QAP objective 275, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



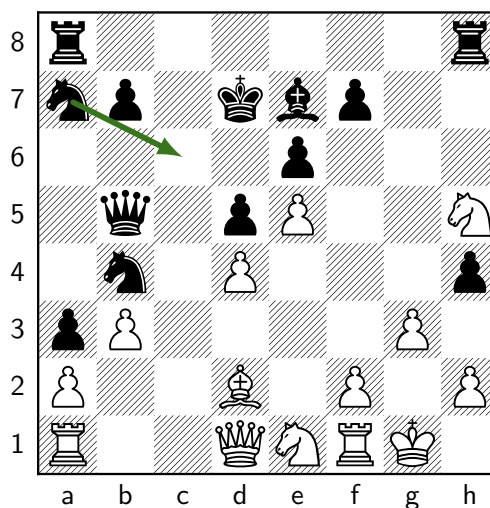
(a)



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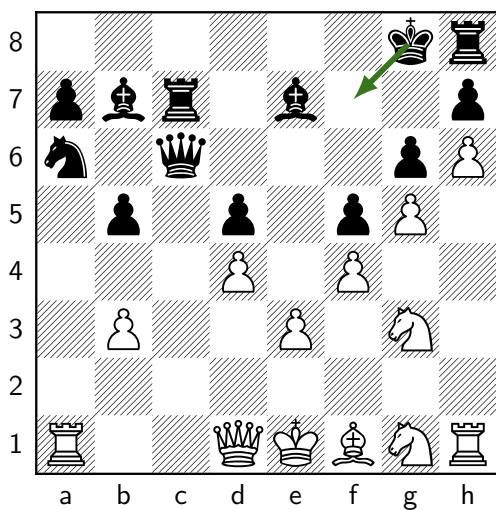


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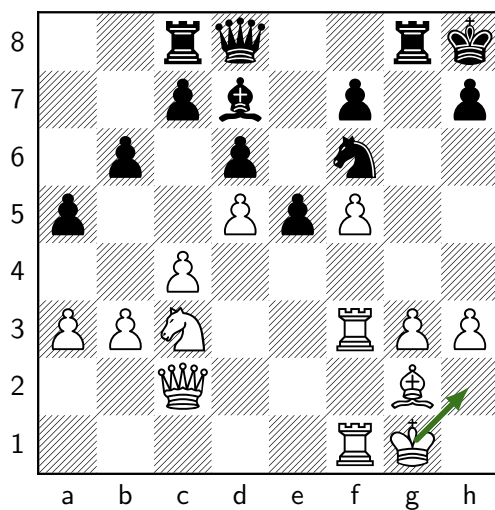


(d)

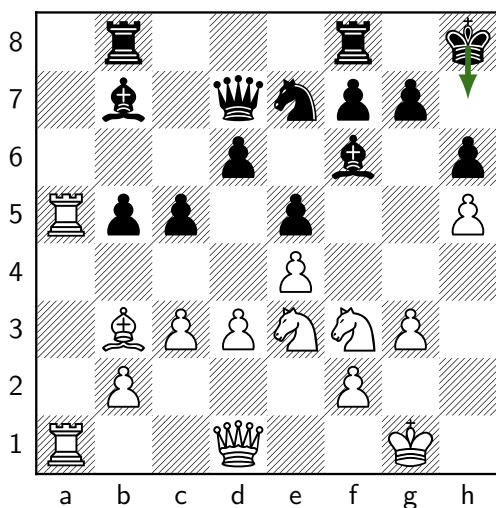
Figure 25: **Move 31, Black: Qc6**. (a) Our position; Melanie plays Qc6 (d7c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#53` (ply 62) [structural distance  $d=85$ , QAP obj=44]: the master played Bf3; under  $\mu$  our Qd2 plays the role of that game’s Bd1 (a **cross-type** correspondence); cross-type roles: our Pd5 as their Rb1, our Pa7 as their Be1; transferred move d7c6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#247` (ply 47) [structural distance  $d=110$ , QAP obj=20]: the master played Ne5; under  $\mu$  our Qd2 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our Be7 as their Qd8, our rh1 as their Bd7; transferred move d7c6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#252` (ply 45) [structural distance  $d=110$ , QAP obj=17]: the master played Nac6; under  $\mu$  our Qd2 plays the role of that game’s Na2 (a **cross-type** correspondence); cross-type roles: our Pb5 as their Ra8, our rh1 as their Pb7; transferred move d7c6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qc6 (summed QAP objective 82, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



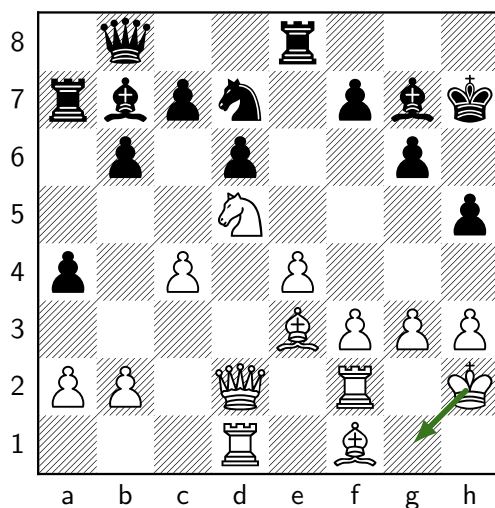
(a)



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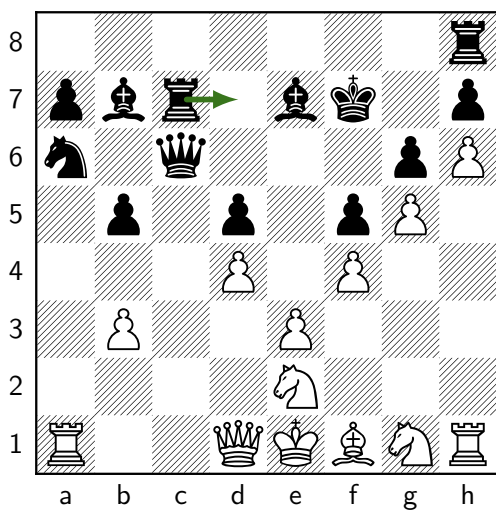


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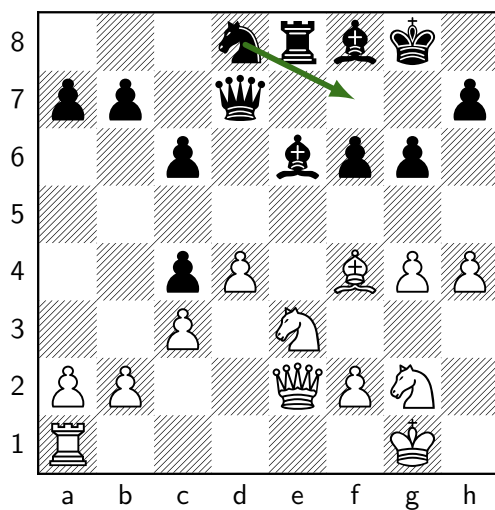


(d)

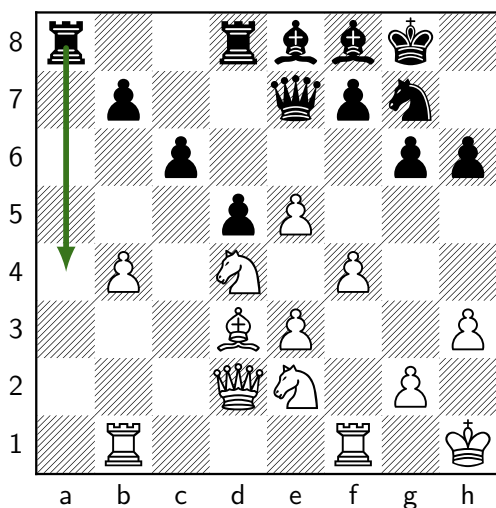
Figure 26: **Move 32, Black: Kf7**. (a) Our position; Melanie plays Kf7 (g8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#814` (ply 52) [structural distance  $d=93$ , QAP obj=60]: the master played Kh2; under  $\mu$  our Kg1 plays the role of that game's Kg1; cross-type roles: our Pg6 as their Qc2, our rh1 as their Bg2; transferred move g8f7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1018` (ply 69) [structural distance  $d=92$ , QAP obj=39]: the master played Kh7; under  $\mu$  our Kg1 plays the role of that game's Kh1; cross-type roles: our Ph7 as their Rf8, our pb3 as their Qd7; transferred move g8f7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2280` (ply 74) [structural distance  $d=86$ , QAP obj=32]: the master played Kg1; under  $\mu$  our Kg1 plays the role of that game's Kh2; cross-type roles: our Ph7 as their Bf1, our Bb7 as their Pa2; transferred move g8f7 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kf7 (summed QAP objective 169, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



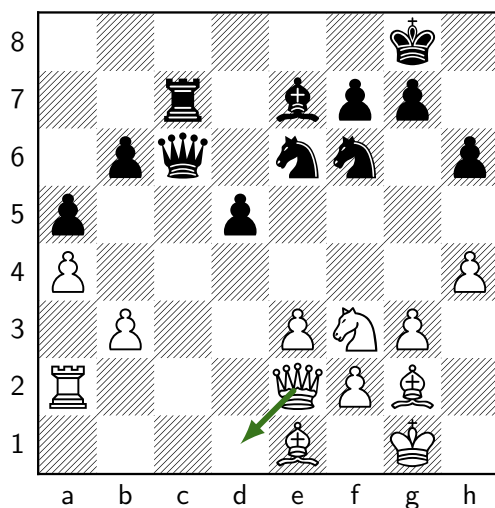
(a)



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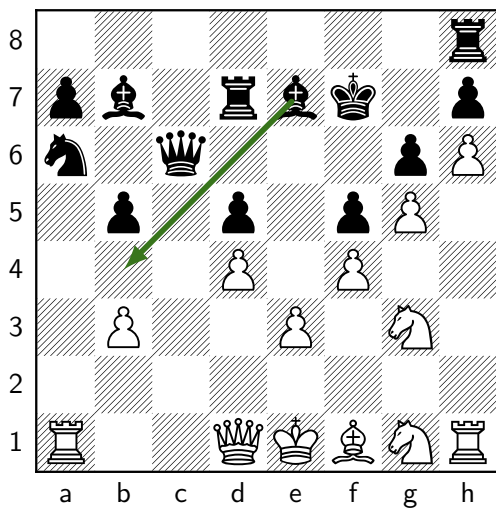


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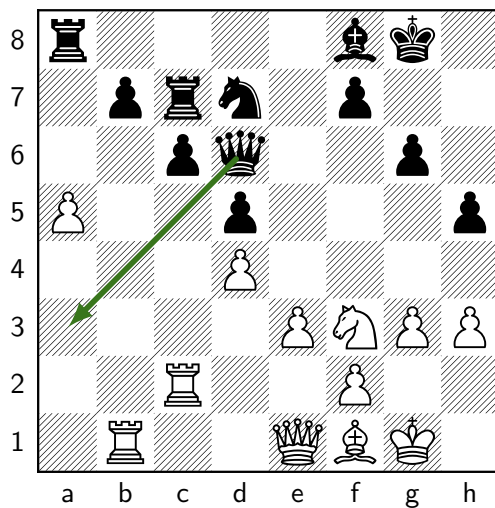


(d)

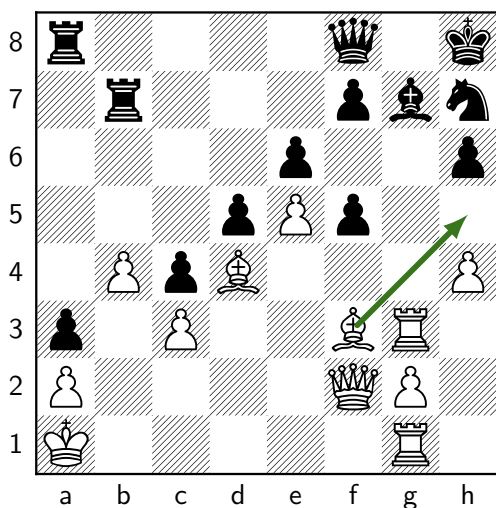
Figure 27: **Move 33, Black: Rd7**. (a) Our position; Melanie plays Rd7 (c7d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1065` (ply 51) [structural distance  $d=110$ , QAP obj=47]: the master played Nf7; under  $\mu$  our Rc2 plays the role of that game’s Nd1 (a **cross-type** correspondence); cross-type roles: our Pd5 as their Re8, our pf4 as their Bf8; transferred move c7d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1764` (ply 55) [structural distance  $d=127$ , QAP obj=42]: the master played Ra4; under  $\mu$  our Rc2 plays the role of that game’s Ra1; cross-type roles: our ne2 as their Rd8, our pf4 as their Be8; transferred move c7d7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3209` (ply 66) [structural distance  $d=116$ , QAP obj=31]: the master played Qd1; under  $\mu$  our Rc2 plays the role of that game’s Qe2 (a **cross-type** correspondence); cross-type roles: our Qc6 as their Ra2, our Pa7 as their Bg2; transferred move c7d7 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Rd7 (summed QAP objective 177, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



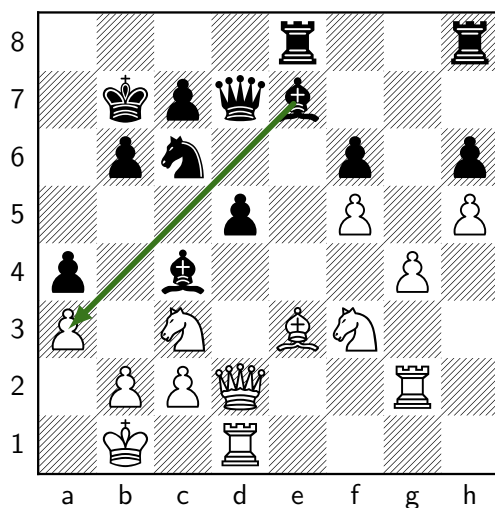
(a)



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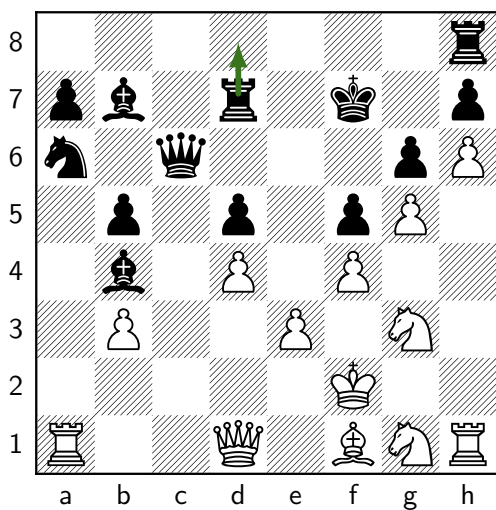


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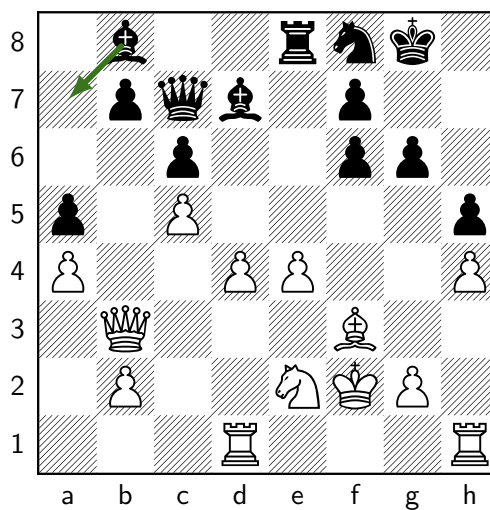


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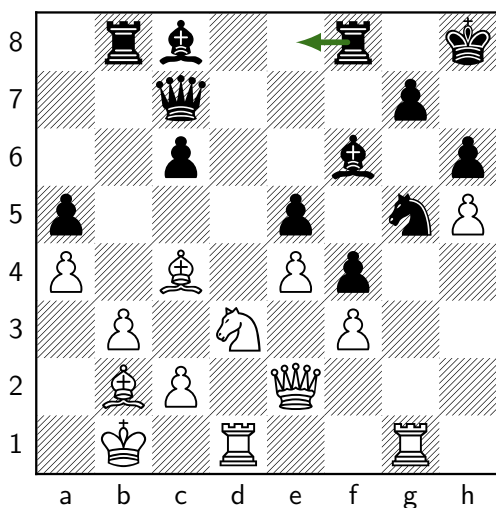
Figure 28: **Move 34, Black: Bb4+.** (a) Our position; Melanie plays Bb4+ (e7b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1106` (ply 57) [structural distance  $d=102$ , QAP obj=55]: the master played Qa3; under  $\mu$  our *Be2* plays the role of that game’s *Qd3* (a **cross-type** correspondence); cross-type roles: our *ng3* as their *Ra8*, our *rh1* as their *Bf8*; transferred move e7b4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#567` (ply 76) [structural distance  $d=106$ , QAP obj=48]: the master played Bh5; under  $\mu$  our *Be2* plays the role of that game’s *Bf3*; cross-type roles: our *Rd7* as their *Qf2*, our *Na6* as their *Rg3*; transferred move e7b4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#2638` (ply 59) [structural distance  $d=103$ , QAP obj=76]: the master played Bxa3; under  $\mu$  our *Be2* plays the role of that game’s *Be2*; cross-type roles: our *Pd5* as their *Re8*, our *Pg6* as their *Rh8*; transferred move e7g5. Synthesis: 2 retrieved analogs transferred to Bb4+ (summed QAP objective 104, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



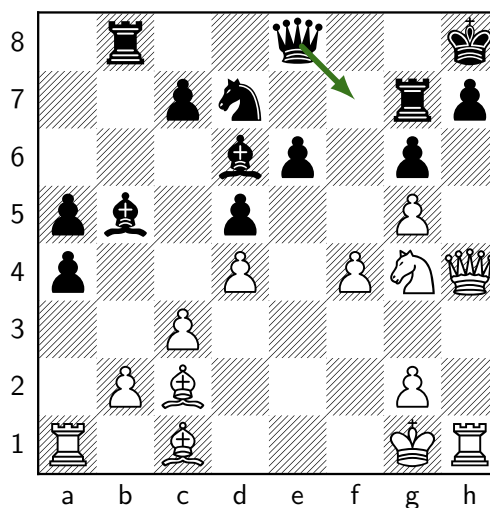
(a)



(b)



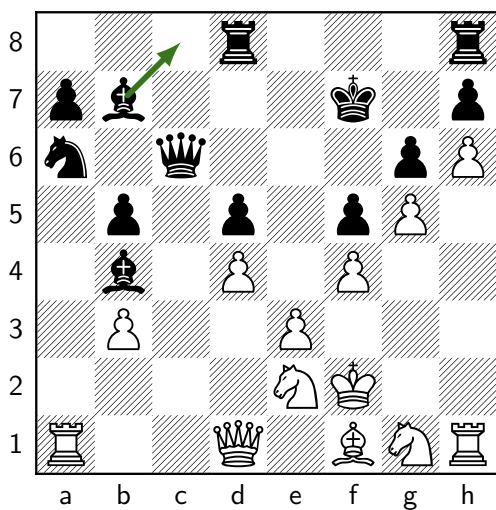
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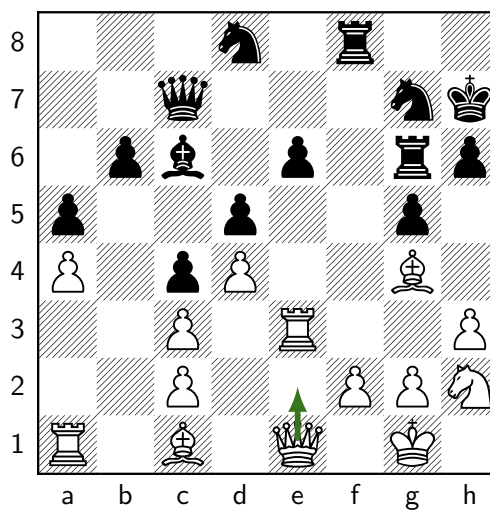
(d)

Figure 29: **Move 35, Black: Rdd8.** (a) Our position; Melanie plays Rdd8 (d7d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1382` (ply 53) [structural distance  $d=114$ , QAP obj=79]: the master played Ba7; under  $\mu$  our *Rd2* plays the role of that game’s *Bb1* (a **cross-type** correspondence); cross-type roles: our *Pd5* as their *Nf8*, our *pd4* as their *Bd7*; transferred move d7d8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#537` (ply 55) [structural distance  $d=105$ , QAP obj=72]: the master played Re8; under  $\mu$  our *Rd2* plays the role of that game’s *Rf1*; cross-type roles: our *bf1* as their *Rb8*, our *pd4* as their *Bc8*; transferred move d7d8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#105` (ply 57) [structural distance  $d=115$ , QAP obj=68]: the master played Qf7; under  $\mu$  our *Rd2* plays the role of that game’s *Qe1* (a **cross-type** correspondence); cross-type roles: our *pf4* as their *Rb8*, our *rh1* as their *Pc7*; transferred move d7d8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rdd8 (summed QAP objective 283, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

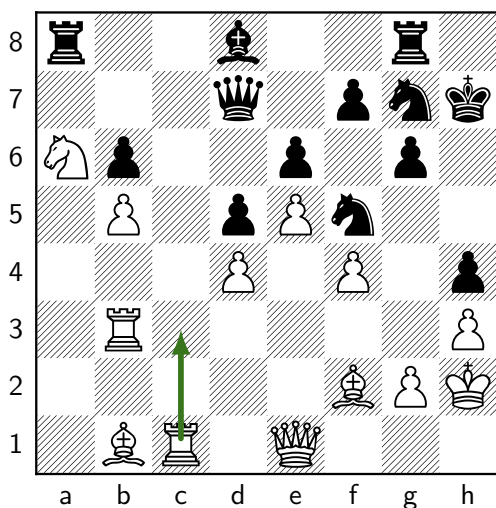




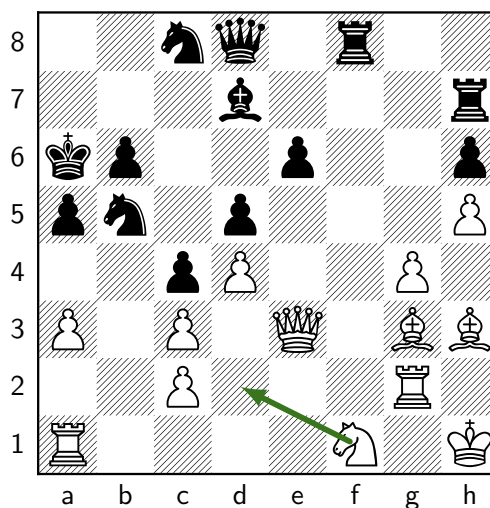
(a)



(b)



(c)



(d)

Figure 30: **Move 36, Black: Bc8.** (a) Our position; Melanie plays Bc8 (b7c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1769` (ply 64) [structural distance  $d=123$ , QAP obj=35]: the master played Qe2; under  $\mu$  our Bb2 plays the role of that game’s Qe1 (a **cross-type** correspondence); cross-type roles: our pf4 as their Ra1, our Pb5 as their Bc1; transferred move b7c8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#627` (ply 78) [structural distance  $d=117$ , QAP obj=14]: the master played Rcc3; under  $\mu$  our Bb2 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our ra1 as their Bb1, our Ph7 as their Bf2; transferred move b7c8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#252` (ply 82) [structural distance  $d=126$ , QAP obj=51]: the master played Nd2; under  $\mu$  our Na3 plays the role of that game’s Nf1; cross-type roles: our Pb5 as their Ra1, our rh1 as their Pc2; transferred move a6c5. Synthesis: 2 retrieved analogs transferred to Bc8 (summed QAP objective 49, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

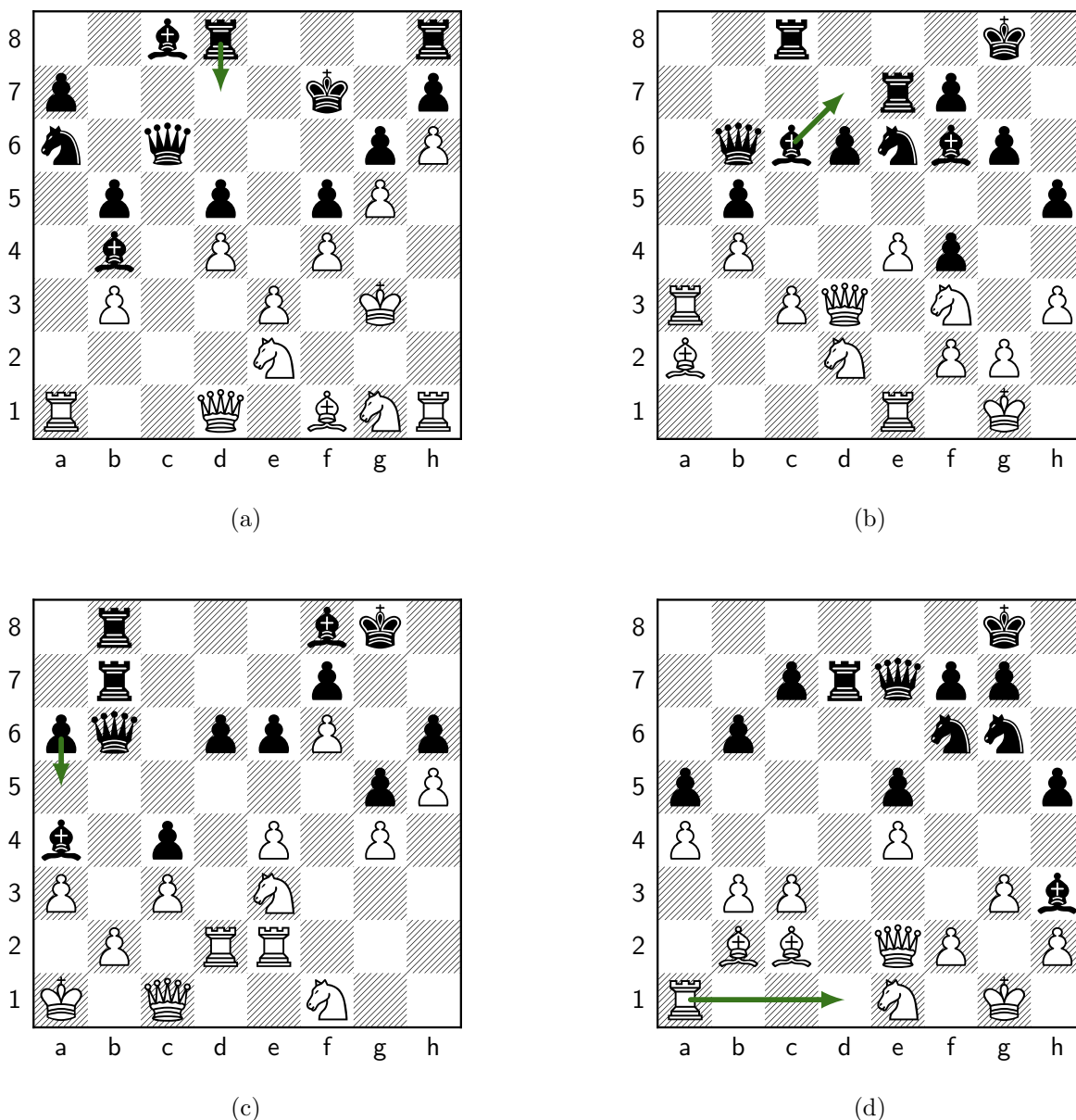
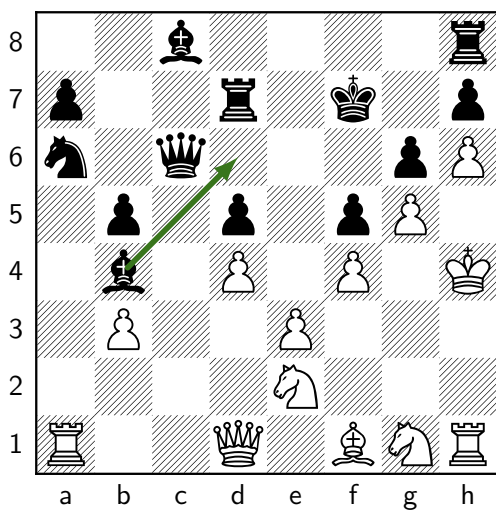
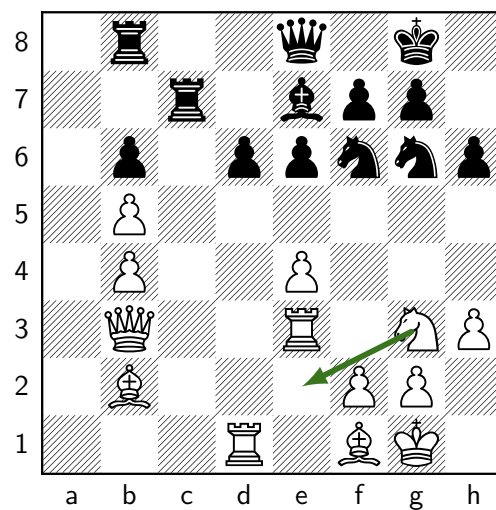


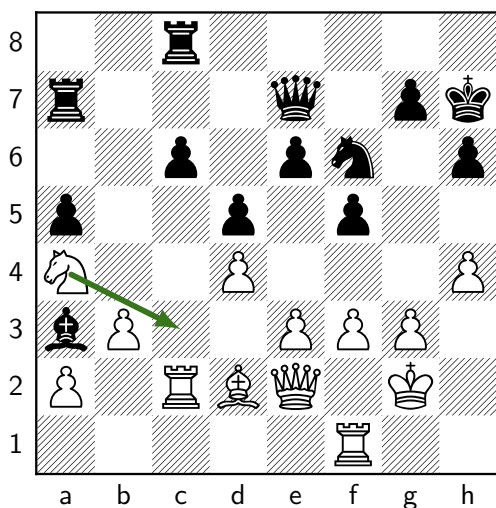
Figure 31: **Move 37, Black: Rd7.** (a) Our position; Melanie plays Rd7 (d8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1055` (ply 63) [structural distance  $d=130$ , QAP obj=28]: the master played Bd7; under  $\mu$  our *Rd1* plays the role of that game’s *Bc3* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Rc8*, our *Qc6* as their *Re7*; transferred move d8d7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1678` (ply 73) [structural distance  $d=116$ , QAP obj=23]: the master played a5; under  $\mu$  our *Rd1* plays the role of that game’s *Pa3* (a **cross-type** correspondence); cross-type roles: our *pg5* as their *Bf8*, our *qd1* as their *Rb7*; transferred move d8d7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2192` (ply 62) [structural distance  $d=123$ , QAP obj=22]: the master played Rd1; under  $\mu$  our *Rd1* plays the role of that game’s *Ra1*; cross-type roles: our *Pb5* as their *Bb2*, our *Qc6* as their *Bc2*; transferred move d8d7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rd7 (summed QAP objective 94, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



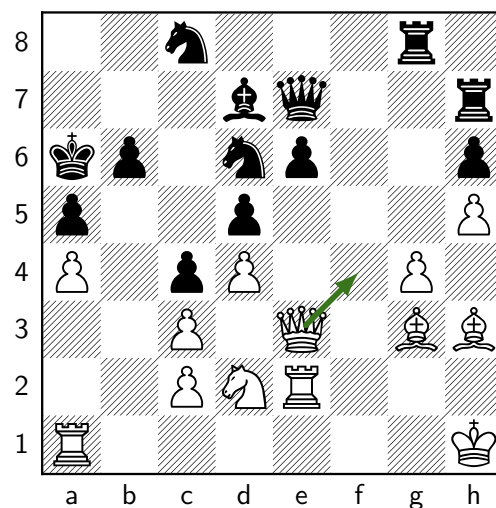
(a)



(b)



(c)



(d)

Figure 32: **Move 38, Black: Bd6**. (a) Our position; Melanie plays Bd6 (b4d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1137` (ply 82) [structural distance  $d=120$ , QAP obj=22]: the master played Ne2; under  $\mu$  our Bb5 plays the role of that game’s Ng3 (a **cross-type** correspondence); cross-type roles: our pd4 as their Rd1, our pf4 as their Bf1; transferred move b4d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1534` (ply 72) [structural distance  $d=129$ , QAP obj=18]: the master played Nc3; under  $\mu$  our Bb5 plays the role of that game’s Na4 (a **cross-type** correspondence); cross-type roles: our ph6 as their Rf1, our bf1 as their Pa2; transferred move b4d6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#252` (ply 88) [structural distance  $d=121$ , QAP obj=18]: the master played Qf4; under  $\mu$  our Bb5 plays the role of that game’s Qe3 (a **cross-type** correspondence); cross-type roles: our ral as their Nd2, our qd1 as their Re2; transferred move b4d6 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bd6 (summed QAP objective 78, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.

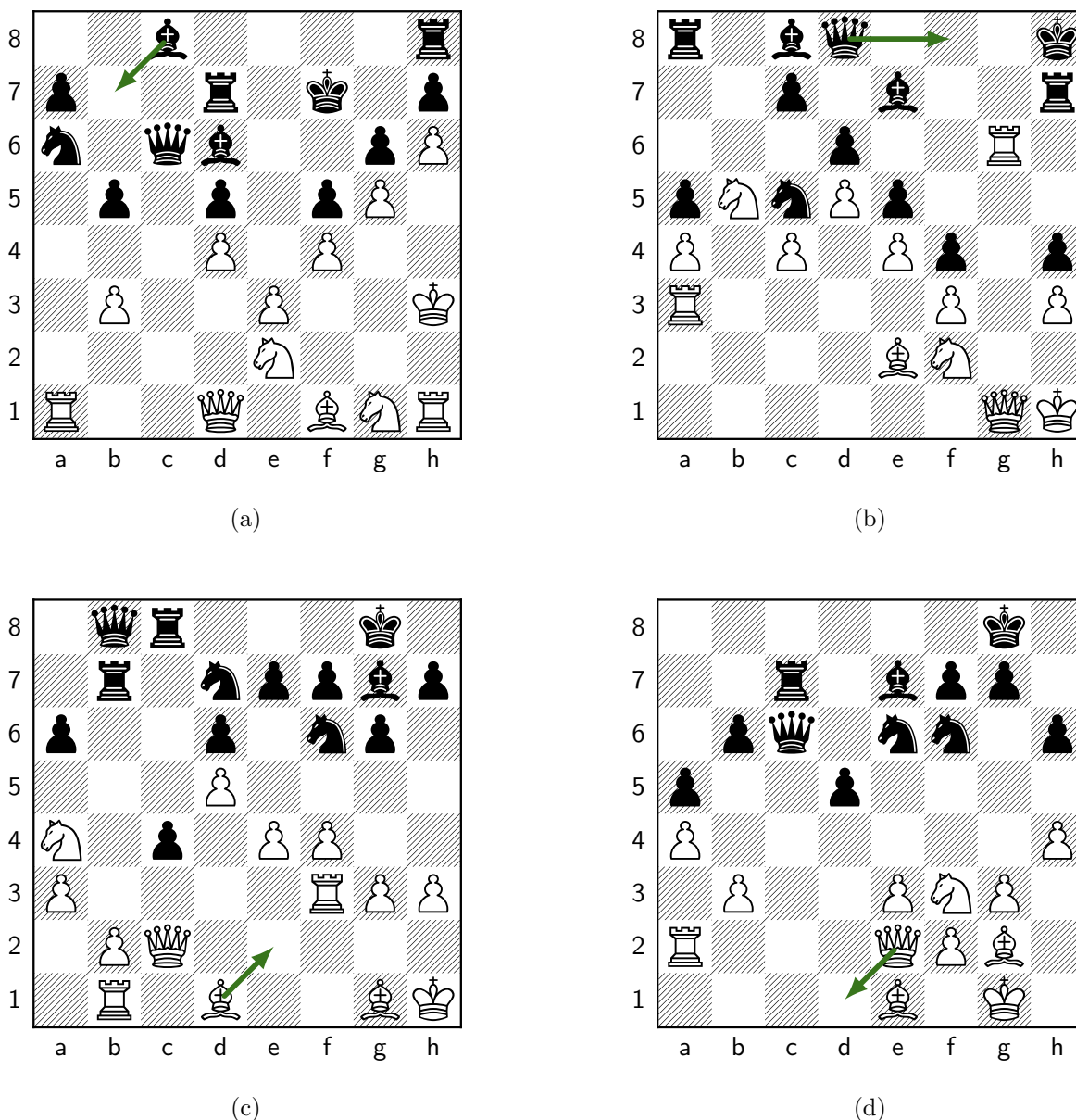
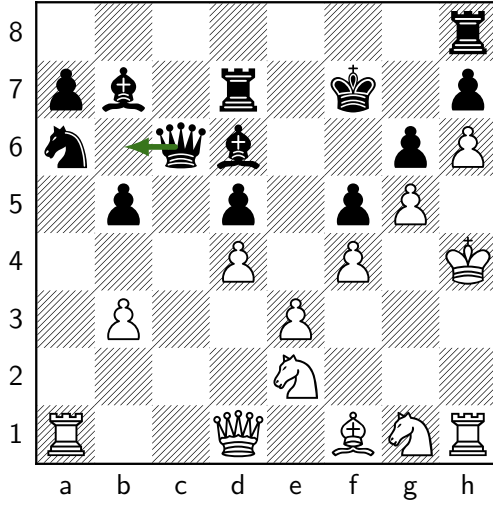
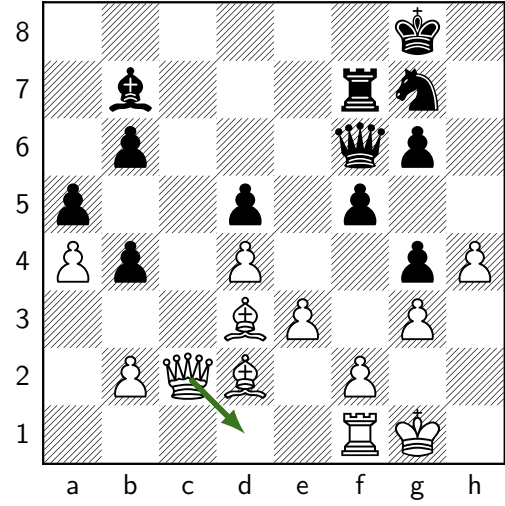


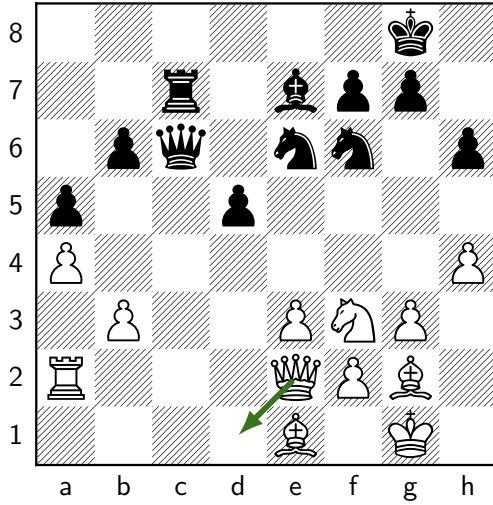
Figure 33: **Move 39, Black: Bb7**. (a) Our position; Melanie plays Bb7 (c8b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#947` (ply 61) [structural distance  $d=140$ , QAP obj=70]: the master played Qf8; under  $\mu$  our Bc1 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our ng1 as their Ra8, our Pd5 as their Bc8; transferred move c8b7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#53` (ply 86) [structural distance  $d=132$ , QAP obj=45]: the master played Be2; under  $\mu$  our Bc1 plays the role of that game’s Bd1; cross-type roles: our qd1 as their Rb1, our Rh8 as their Bg1; transferred move c8b7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3209` (ply 66) [structural distance  $d=141$ , QAP obj=34]: the master played Qd1; under  $\mu$  our Bc1 plays the role of that game’s Qe2 (a **cross-type** correspondence); cross-type roles: our Qc6 as their Ra2, our Pa7 as their Bg2; transferred move c8b7 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Bb7 (summed QAP objective 262, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



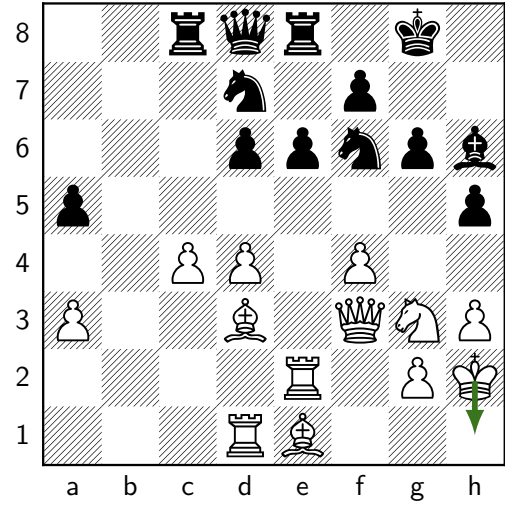
(a)



(b)

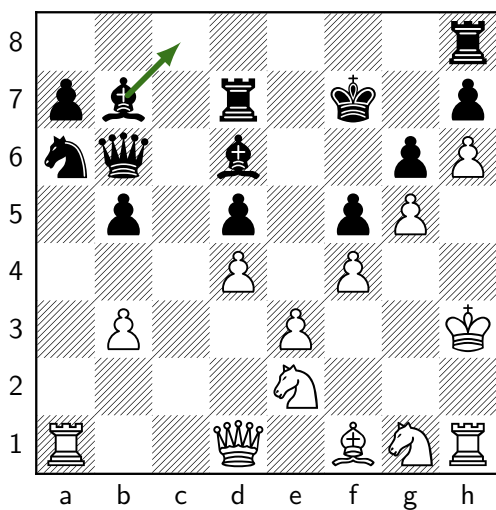


(c)

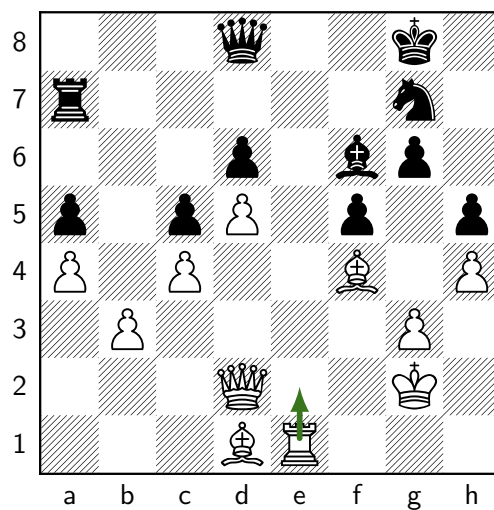


(d)

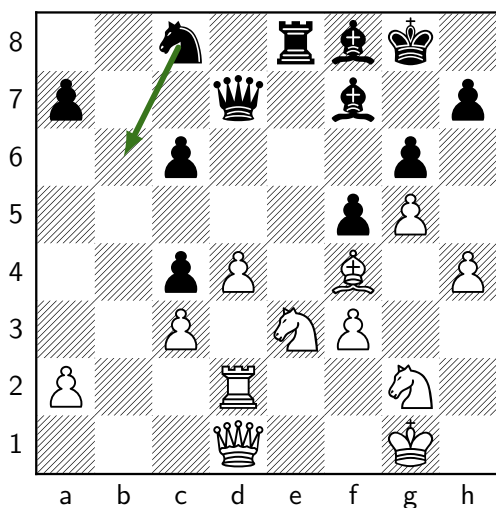
Figure 34: **Move 40, Black: Qb6.** (a) Our position; Melanie plays Qb6 (c6b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2163` (ply 86) [structural distance  $d=129$ , QAP obj=61]: the master played Qd1; under  $\mu$  our Qc3 plays the role of that game’s Qc2; cross-type roles: our Na6 as their Pd4, our Na6 as their Ph4; transferred move c6b6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3209` (ply 66) [structural distance  $d=117$ , QAP obj=34]: the master played Qd1; under  $\mu$  our Qc3 plays the role of that game’s Qe2; cross-type roles: our Bb7 as their Ra2, our Pa7 as their Bg2; transferred move c6b6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3419` (ply 62) [structural distance  $d=130$ , QAP obj=50]: the master played Kh1; under  $\mu$  our Kf2 plays the role of that game’s Kh2; cross-type roles: our Qc6 as their Rd1, our Rd7 as their Be1; transferred move f7e8. Synthesis: 3 retrieved analogs transferred to Qb6 (summed QAP objective 155, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



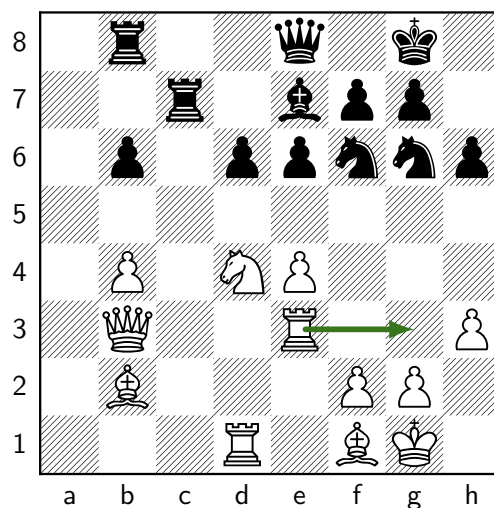
(a)



(b)

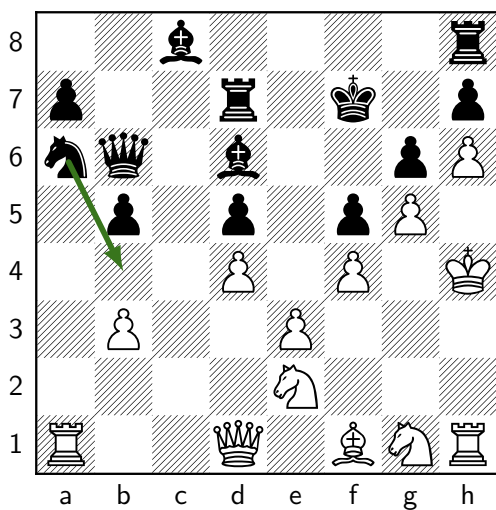


(c)

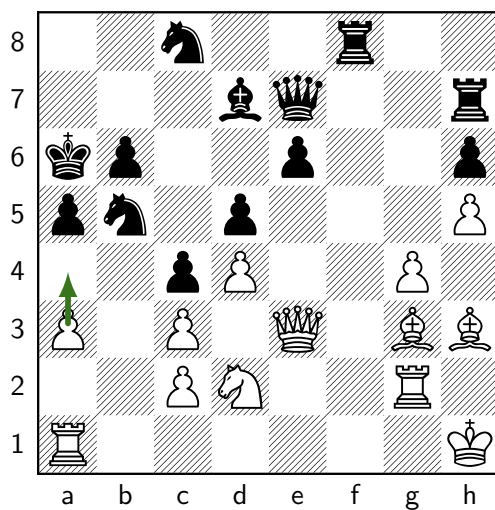


(d)

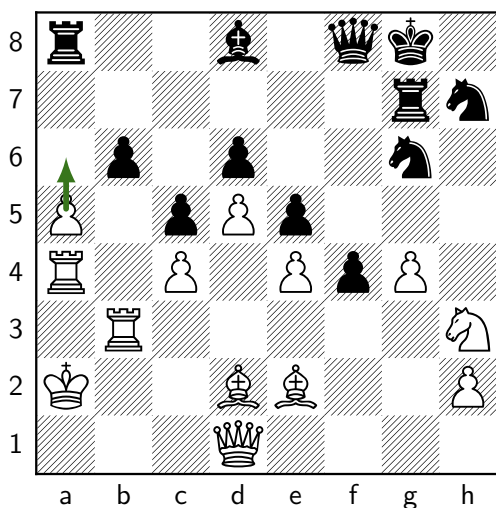
Figure 35: **Move 41, Black: Bc8.** (a) Our position; Melanie plays Bc8 (b7c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2662` (ply 88) [structural distance  $d=129$ , QAP obj=52]: the master played Re2; under  $\mu$  our Bb2 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our Pb5 as their Bd1, our Rd7 as their Pg3; transferred move b7c8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1065` (ply 71) [structural distance  $d=126$ , QAP obj=48]: the master played Nb6; under  $\mu$  our Bb2 plays the role of that game’s Nc1 (a **cross-type** correspondence); cross-type roles: our Bd6 as their Re8, our ph6 as their Bf8; transferred move b7c8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1137` (ply 86) [structural distance  $d=139$ , QAP obj=46]: the master played Rg3; under  $\mu$  our Bb2 plays the role of that game’s Re3 (a **cross-type** correspondence); cross-type roles: our Pb5 as their Rd1, our Pd5 as their Bf1; transferred move b7c8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bc8 (summed QAP objective 205, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



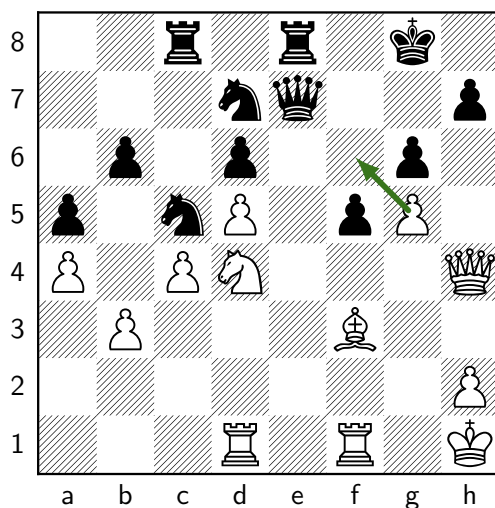
(a)



(b)



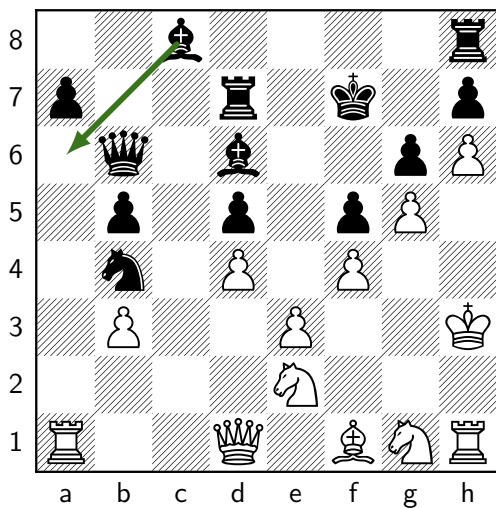
(c)



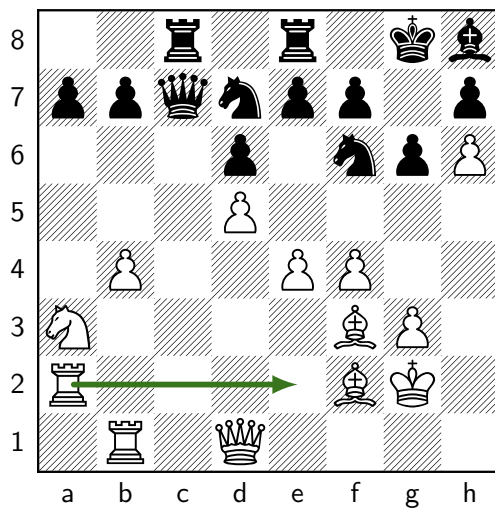
(d)

Figure 36: **Move 42, Black: Nb4**. (a) Our position; Melanie plays Nb4 (a6b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#252` (ply 84) [structural distance  $d=108$ , QAP obj=86]: the master played a4; under  $\mu$  our Na3 plays the role of that game's Pa3 (a **cross-type** correspondence); cross-type roles: our Bd6 as their Nd2, our pg5 as their Rg2; transferred move a6b4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2984` (ply 96) [structural distance  $d=106$ , QAP obj=36]: the master played a6; under  $\mu$  our Na3 plays the role of that game's Pa5 (a **cross-type** correspondence); cross-type roles: our Pa7 as their Qd1, our Qb6 as their Bd2; transferred move a6b4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3357` (ply 68) [structural distance  $d=114$ , QAP obj=67]: the master played gxf6; under  $\mu$  our Na3 plays the role of that game's Pg5 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Rd1, our Ph7 as their Bf3; transferred move a6c5. Synthesis: 2 retrieved analogs transferred to Nb4 (summed QAP objective 122, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.

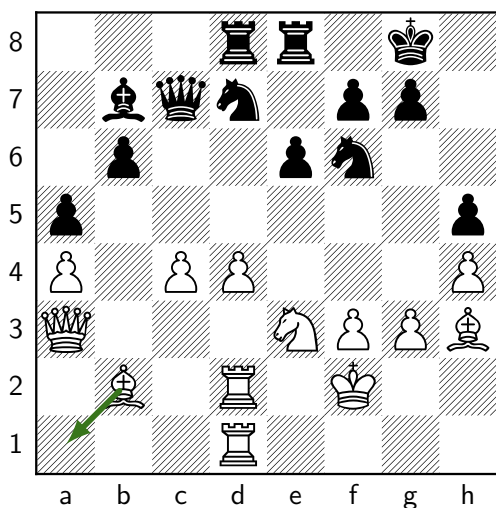




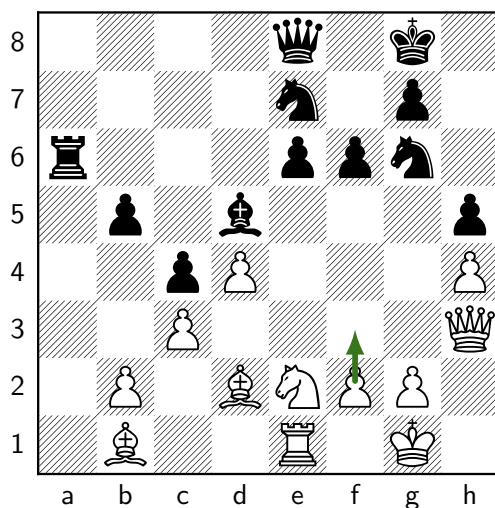
(a)



(b)

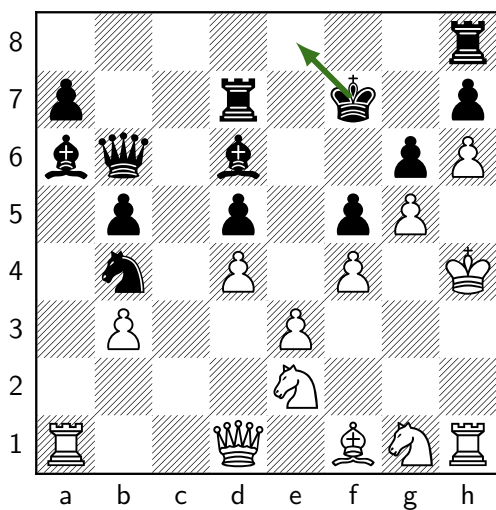


(c)

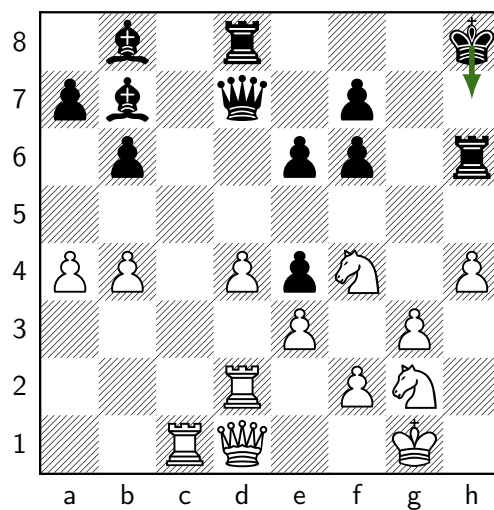


(d)

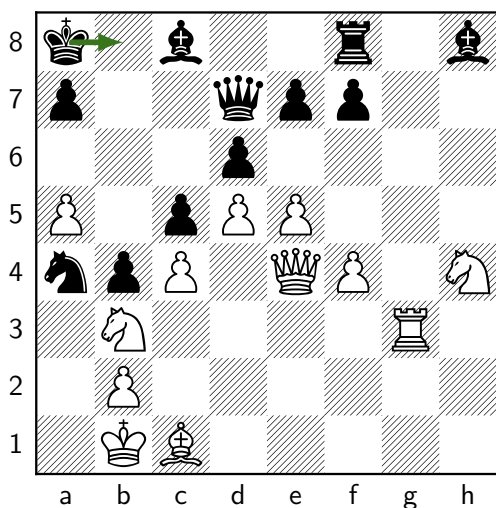
Figure 37: **Move 43, Black: Ba6**. (a) Our position; Melanie plays Ba6 (c8a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1038` (ply 84) [structural distance  $d=124$ , QAP obj=47]: the master played Re2; under  $\mu$  our Bc1 plays the role of that game’s Ra2 (a **cross-type** correspondence); cross-type roles: our Pd5 as their Rb1, our ne2 as their Qd1; transferred move c8a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2789` (ply 80) [structural distance  $d=128$ , QAP obj=47]: the master played Ba1; under  $\mu$  our Bc1 plays the role of that game’s Bb2; cross-type roles: our bf1 as their Rd1, our ne2 as their Qa3; transferred move c8a6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2364` (ply 74) [structural distance  $d=104$ , QAP obj=44]: the master played f3; under  $\mu$  our Bc1 plays the role of that game’s Pf2 (a **cross-type** correspondence); cross-type roles: our Pg6 as their Bd2, our Pf5 as their Ne2; transferred move c8a6 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ba6 (summed QAP objective 193, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



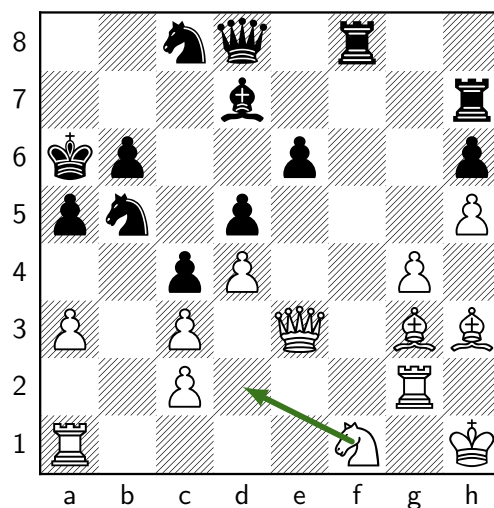
(a)



(b)

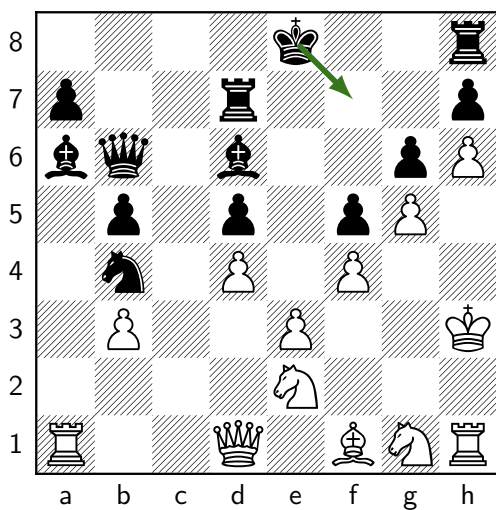


(c)

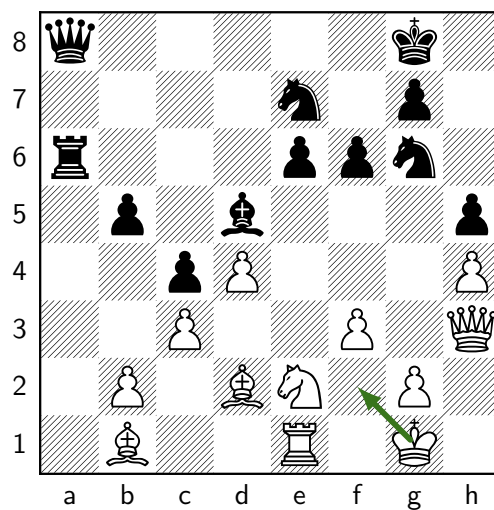


(d)

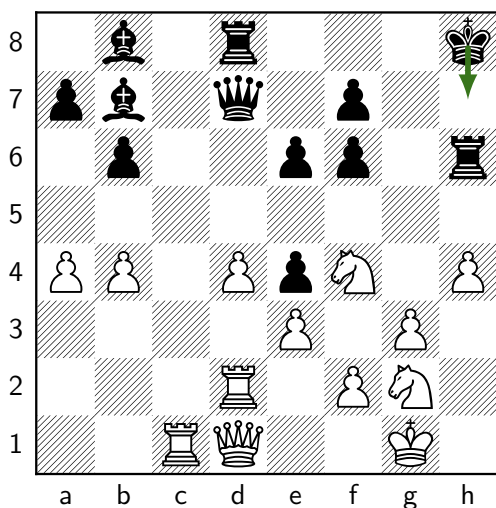
Figure 38: **Move 44, Black: Ke8 (f7e8).** (a) Our position; Melanie plays Ke8 (f7e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#133` (ply 75) [structural distance  $d=105$ , QAP obj=50]: the master played Kh7; under  $\mu$  our Kf2 plays the role of that game’s Kh1; cross-type roles: our Ra1 as their Bb8, our Qd1 as their Pa7; transferred move f7e8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3656` (ply 85) [structural distance  $d=117$ , QAP obj=37]: the master played Kb8; under  $\mu$  our Kf2 plays the role of that game’s Ka1; cross-type roles: our Rh8 as their Bc8, our Qd1 as their Rf8; transferred move f7e8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#252` (ply 82) [structural distance  $d=115$ , QAP obj=78]: the master played Nd2; under  $\mu$  our Ba3 plays the role of that game’s Nf1 (a **cross-type** correspondence); cross-type roles: our Nb4 as their Ra1, our Pa7 as their Rg2; transferred move a6c8. Synthesis: 6 retrieved analogs transferred to Ke8 (summed QAP objective 232, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



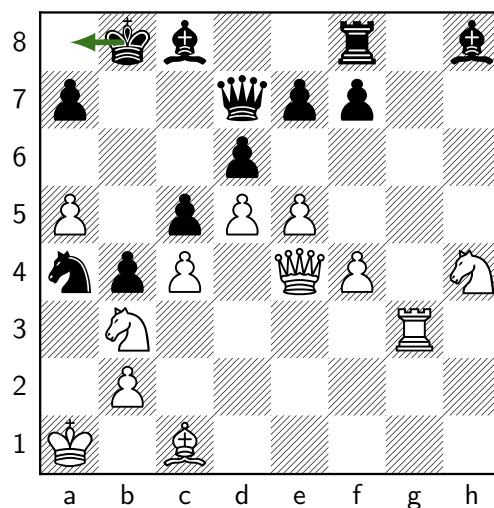
(a)



(b)

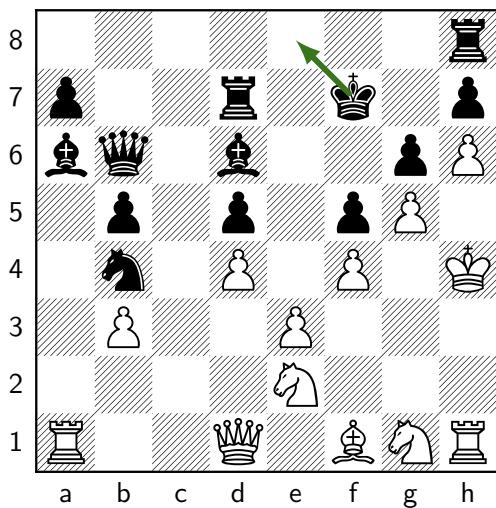


(c)

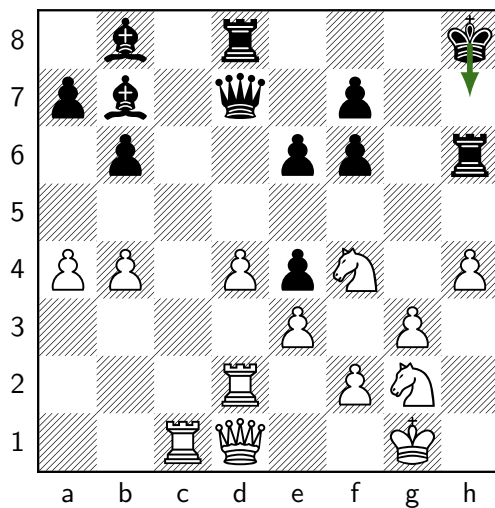


(d)

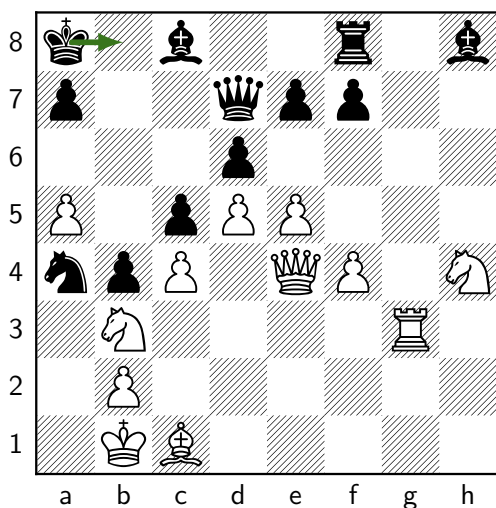
Figure 39: **Move 45, Black: Kf7 (e8f7).** (a) Our position; Melanie plays Kf7 (e8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2364` (ply 76) [structural distance  $d=129$ , QAP obj=53]: the master played Kf2; under  $\mu$  our *Ke1* plays the role of that game’s *Kg1*; cross-type roles: our *ra1* as their *Bb1*, our *qd1* as their *Re1*; transferred move e8f7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#133` (ply 75) [structural distance  $d=123$ , QAP obj=46]: the master played Kh7; under  $\mu$  our *Ke1* plays the role of that game’s *Kh1*; cross-type roles: our *ra1* as their *Bb8*, our *qd1* as their *Rd8*; transferred move e8f7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3656` (ply 83) [structural distance  $d=141$ , QAP obj=36]: the master played Ka8; under  $\mu$  our *Ke1* plays the role of that game’s *Kb1*; cross-type roles: our *Pg6* as their *Bc8*, our *qd1* as their *Rf8*; transferred move e8f7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kf7 (summed QAP objective 161, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



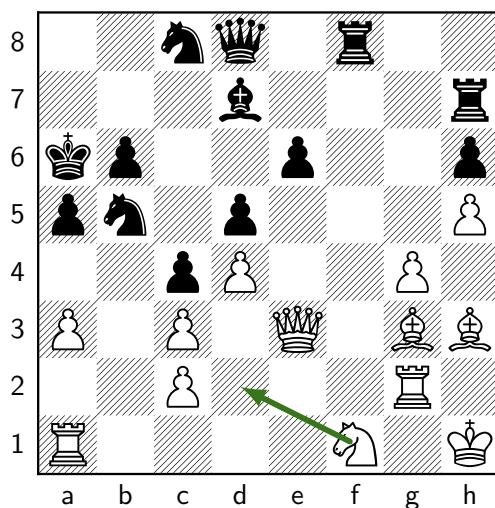
(a)



(b)



(c)



(d)

Figure 40: **Move 46, Black: Ke8 (f7e8).** (a) Our position; Melanie plays Ke8 (f7e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#133` (ply 75) [structural distance  $d=105$ , QAP obj=50]: the master played Kh7; under  $\mu$  our Kf2 plays the role of that game’s Kh1; cross-type roles: our Ra1 as their Bb8, our Qd1 as their Pa7; transferred move f7e8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3656` (ply 85) [structural distance  $d=117$ , QAP obj=37]: the master played Kb8; under  $\mu$  our Kf2 plays the role of that game’s Ka1; cross-type roles: our Rh8 as their Bc8, our Qd1 as their Rf8; transferred move f7e8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#252` (ply 82) [structural distance  $d=115$ , QAP obj=78]: the master played Nd2; under  $\mu$  our Ba3 plays the role of that game’s Nf1 (a **cross-type** correspondence); cross-type roles: our Nb4 as their Ra1, our Pa7 as their Rg2; transferred move a6c8. Synthesis: 6 retrieved analogs transferred to Ke8 (summed QAP objective 232, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.